Thesis Notes

Nathaniel Kenschaft

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Contents

0.1 TODO

1. ~~Read [Far06]~~
2. ~~Read [CG13]~~
3. Read [DLS24]
4. ~~Ok so there's no continuous section in [DLS24], but what is the Schwarz genus? I.e. how many cases are there?~~
5. ~~Is it a good idea to just canonicalize via $\mathbb{S}_d^n$? Why or why not?~~
6. ~~What can we learn about $X = \left\{A \in \mathbb{R}^{d\times n} : \operatorname{rank}(A) = d\right\}$ and $X/O(d)$ or $X/SO(d)$?~~
7. Is a canonicalization up to homotopy sufficient?
8. ~~Use restrictions to bound sectional category on original space~~
9. ~~Consider $\mathrm{SO}(d)$ acting on $\mathbb{R}^{d\times n}$~~
10. ~~Understand proof that there is no canonicalization of $(\mathbb{R}^{d\times n}, \mathrm{O}(d))$ and $(\mathbb{R}^{d\times n}, \mathrm{SO}(d))$~~.
11. Experiment (computationally) with patchwork continuous sections
12. Consider what group acts on vectorized images and consider Schwarz genus
13. What if we want a *differentiable* section?
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15. What can we say about the cardinality of continuity-preserving frames for $\mathrm{O}(d)$ and $\mathrm{SO}(d)$ for $d > 2$?

Chapter 1

Introduction

1.1 Motivation

In machine learning, one is generally trying to learn a map $f : V \to W$ between vector spaces with some desired properties. For instance, if V is the vector space of tuples of age and height of some population and W represents the weights of that population, one may wish to identify a relationship between those three variables in the form of a map f taking tuples $(age, height) \in V$ to some $weight = f(age, height) \in W$. Generally, we want this map to reflect our intuition of how patterns in nature behave, meaning this map should at the very least be continuous (or nearly so), and potentially respect other properties or structure of the data.

A natural example of additional structure is when the data have some sort of inherent symmetry. This could arise when trying to label the pose of a person, for example, where any horizontal rotation of a person in a certain pose still is shaped in that pose. Mathematically, we can represent this with a group action on our space V by some group G of symmetries. In the case of poses, we might wish to construct a map f that ignores the actions our group G of horizontal rotations, meaning that every rotation is identified as the same pose. Formally, this is called an *invariant* or *G-invariant* map, defined to mean that $f(g \cdot v) = f(v)$ for all $g \in G$ and $v \in V$. Similarly, one may also wish to consider the case where W is also equipped with a G-action (for the same group G), in which case we may want a *(G-)equivariant* map $f : V \to W$, meaning $f(g \cdot v) = g \cdot f(v)$.

For practical purposes, one must then consider how to construct such a map. A very natural way is to consider a so-called *canonicalization*, where one first associates each instance of the data to a canonical representative respecting the symmetry [DLS24]. Concretely, if someone hands you a collection of points representing the joints of a human in a certain pose, you may attempt to rotate this collection of points so that the person would be facing you, then proceed to identify the pose. Mathematically, a canonicalization is then a map $c : V \to V$ such that $c(g \cdot v) = c(v) \in Gv$ for all $g \in G$ and $v \in V$. Then any continuous map $f : V \to W$, not necessarily invariant on its own, becomes an invariant map when composed with c, since $f(c(g \cdot v)) = f(c(v))$. The question then becomes whether or not such a canonicalization even exists. This question is addressed in [DLS24] and chapter 2.

Another rather intuitive way to construct such a map would be to take an average of some map over all possible symmetries. Informally, this would look like

$$\frac{1}{\operatorname{card}(G)} \sum_{g \in G} f(g \cdot v).$$

This of course doesn't make sense if the group G is infinite (like the case of rotational symmetries), but it can be made precise using integration and measure theory; this is explored further in [DLS24] and chapter 3.

1.2 Notation

- I use $A \cong B$ to denote that there exists an isomorphism between A and B in whatever the appropriate category is. For instance if X and Y are topological spaces then $X \cong Y$ means X and Y are homeomorphic, or if G and H are groups then $G \cong H$ means G and H are isomorphic as groups.
- For a homotopy $h : X \times [0,1] \to Y$, I may also write $h_t : X \to Y$ with $h_t(x) := h(x,t)$.
- For any set X, I write c_x to mean the constant function at $x \in X$.
- Given a path $\gamma : [0,1] \to X$, I write $\overline{\gamma}$ to mean the reversed path $\overline{\gamma}(t) := \gamma(1-t)$.
- For any two paths $\alpha, \beta : [0,1] \to X$ with $\alpha(1) = \beta(0)$, I write $\beta * \alpha$ to indicate their concatenation:
$$(\beta * \alpha)(t) = \begin{cases} \alpha(2t) & \text{if } 0 \le t \le \frac{1}{2} \\ \beta(2t-1) & \text{if } \frac{1}{2} \le t \le 1 \end{cases}.$$
The exact paramerization is unimportant, but if it ever matters this is what I mean.
- Unless specified otherwise, for any group G I will write $e \in G$ to be the identity in G. If there is the potential for confusion between different groups, I will use subscripts to differentiate (e.g. e_G is the identity in G, e_H is the identity in H).
- I write $\mathbb{R}^{d\times n}_{\operatorname{rank} k}$ to mean the set $\left\{A \in \mathbb{R}^{d\times n} : \operatorname{rank}(A) = k\right\}$ and set $\mathbb{R}^{d\times n}_{*} = \mathbb{R}^{d\times n}_{\operatorname{rank} d}$.
- Given some matrix $A \in \mathbb{R}^{n\times m}$, I write $A[r_1 : r_2, c_1 : c_2]$ to do indicate the matrix of entries taken from rows r_1 through r_2 (inclusive) and columns c_1 through c_2 (inclusive) of A. If any value is left empty, I mean to include all the way to the first/last row/column depending on which entry is empty (e.g. $A[:, : 2]$ is the first two columns of A). For any programmers, this is just the Python library `numpy`'s syntax.

Chapter 2

Canonicalization

2.1 Background

2.1.1 Background from [Far06]

Definition 2.1. *For fixed $a = (a_1, ..., a_m) \in \mathbb{R}_+^m$, we define the variety*

$$M(a) = \left\{ (z_1, ..., z_m) \in (S^2)^m : \sum_{i=1}^m a_i z_i = 0 \right\} / SO(3)$$

where $SO(3)$ acts coordinate-wise on $(S^2)^m$. This variety describes all polygons in $\mathbb{R}^3$ with sidelengths $a_1, ..., a_m$.

Definition 2.2. *Let X be a path-connected topological space. Then $PX = C([0,1]; X)$ is the space of all paths $\gamma : [0,1] \to X$ equipped with the compact-open topology.*

Recall that the compact-open topology has a subbase given by sets of the form

$$V(K, U) := \{ f \in C(X, Y) : f[K] \subseteq U \}$$

for $K \subseteq [0,1]$ compact and $U \subseteq X$ open [Wik26a].

Definition 2.3. *For spaces E and B, a map $p : E \to B$ has the* homotopy lifting property (HLP) *for a space X if for every homotopy $h_t : X \to B$ and every lift $\widetilde{h}_0 : X \to E$ of h_0 ($h_0 = p \circ \widetilde{h}_0$), there exists a homotopy $\widetilde{h}_t : X \to E$ that lifts h_t ($h_t = p \circ \widetilde{h}_t$).*

Definition 2.4. *For spaces E and B, a map $p : E \to B$ is a* (Hurewicz) fibration *if p satisfies the homotopy lifting property for all topological spaces [Wik26c].*

Remark 2.1. *The map $\pi : PX \to X \times X$ given by $\pi(\gamma) = (\gamma(0), \gamma(1))$ is a fibration.*

Proof. Let Y be any topological space and let $h_t : Y \to X \times X$ be a homotopy with some $\widetilde{h}_0 : Y \to PX$ such that $h_0 = \pi \circ \widetilde{h}_0$. Then we simply define

$$\widetilde{h}_t(y)(s) = \begin{cases} (h_{t(1-3s)}(y))_1 & \text{if } 0 \leq s \leq \frac{1}{3} \\ \widetilde{h}_0(y)(3(s - \frac{1}{3})) & \text{if } \frac{1}{3} \leq s \leq \frac{2}{3} \\ (h_{t(3s-2)}(y))_2 & \text{if } \frac{2}{3} \leq s \leq 1 \end{cases}.$$

Intuitively, this simply defines a path from $(h_t(y))_1$ to $(h_t(y))_2$ by following:

1. $(h_\bullet(y))_1$ from $(h_t(y))_1$ to $(h_0(y))_1$,
2. $\widetilde{h}_0$ from $(h_0(y))_1 = \widetilde{h}_0(y)(0)$ ($h_0 = \pi \circ \widetilde{h}_0$) to $\widetilde{h}_0(y)(1)$,
3. and $(h_\bullet(y))_2$ from $\widetilde{h}_0(y)(1) = (h_0(y))_2$ ($\pi \circ \widetilde{h}_0 = h_0$) to $(h_t(y))_2$.

By construction $\widetilde{h}_t$ is continuous, so we conclude that π is a fibration. $\square$

Definition 2.5. *A* motion planning algorithm *is a section $s : X \times X \to PX$ of π, i.e. $\pi \circ s = \mathrm{id}_{X \times X}$.*

Lemma 2.1. *A continuous motion planning algorithm in X exists if and only if X is contractible.*

Proof. First assume that there exists a continuous motion planning algorithm, i.e. a continuous section $s : X \times X \to PX$ of π, and fix some $x_0 \in X$. Then for every $x \in X$, define $\gamma_x = s(x_0, x) \in PX$. Now define a homotopy $h_t : X \to X$ via $h_t(x) = \gamma_x(t)$. Observe that $h_0(x) = \gamma_x(0) = x_0$ and $h_1(x) = \gamma_x(1) = x$ so $h_0 = c_{x_0}$ and $h_1 = \mathrm{id}_X$. Since s is continuous, this means that h is a homotopy from c_{x_0} to id_X and therefore X is contractible.

Now instead assume that X is contractible and let $h_t : X \to X$ be a homotopy exhibiting this ($h_0 = c_{x_0}$ for some $x_0 \in X$ and $h_1 = \mathrm{id}_X$). Then for each $x \in X$, define $\gamma_x \in PX$ via $\gamma_x(t) = h_t(x)$ and define $s : X \times X \to PX$ by $s(x, y) = \gamma_y * \overline{\gamma_x}$. Then s is a section since $\pi(s(x, y)) = ((\gamma_y * \overline{\gamma_x})(0), (\gamma_y * \overline{\gamma_x})(1)) = (x, y)$, and s is continuous by construction. □

Definition 2.6. *A topological space X is a* Euclidean Neighborhood Retract (ENR) *if there exists an embedding $i : X \to \mathbb{R}^n$ for some n with an open neighborhood U of $i[X]$ such that U retracts onto $i[X]$.*

Definition 2.7. *A motion planning algorithm $s : X \times X \to PX$ is* tame *if there are finitely many disjoint sets $F_i \subseteq X \times X$ such that $X \times X = \cup_i F_i$, $s|_{F_i}$ is continuous, and F_i is an ENR.*

Definition 2.8. *The* topological complexity *of a tame motion planning algorithm is the minimum number of sets F_i one can take in definition 2.7.*

Definition 2.9. *For a path-connected space X, the topological complexity $\mathbf{TC}_1(X)$ is the minimum topological complexity of tame motion planning algorithms in X. We set $\mathbf{TC}_1(X) = \infty$ if X admits no tame motion planning algorithms.*

Remark 2.2. $\mathbf{TC}_1(X) = 1$ *if and only if X is a contractible ENR.*

Proof. This trivially follows from lemma 2.1 and definitions 2.6 and 2.7. □

Definition 2.10. *The* Schwarz genus *or* sectional category *of a fibration $p : E \to B$, denoted* $\mathrm{secat}(p)$ *is the minimum number k of open sets $U_1, ..., U_k$ covering B such that each U_j admits a continuous section $s_j : U_j \to E$ of p.*

Remark 2.3. *The Schwarz genus of a fibration $p : E \to B$ is 1 if and only if p admits a continuous section $s : B \to E$.*

Definition 2.11. *For a path-connected pointed space (X, x_0), we define*

$$P_0X := \{\gamma \in C([0, 1]; X) : \gamma(0) = x_0\}.$$

Definition 2.12. *For a path-connected pointed space (X, x_0), the* Lusternik-Schnirelmann category of X*, denoted* $\mathrm{cat}(X)$*, is the Schwarz genus of the fibration $\pi_0 : P_0X \to X, \gamma \mapsto \gamma(1)$.*

Definition 2.13. *For a path-connected space X, the topological complextiy $\mathbf{TC}_2(X)$ is the Schwarz genus of the fibration $\pi : PX \to X \times X$.*

Remark 2.4. $\mathbf{TC}_2(X) = 1$ *if and only if X is contractible.*

Proof. This is a direct corollary of lemma 2.1. □

Lemma 2.2. *Let $p : E \to B$ be a fibration, and let $B' \subseteq B$ with $E' := p^{-1}[B']$ and $p' = p|_{E'} : E' \to B'$. Then* $\mathrm{secat}(p') \leq \mathrm{secat}(p)$.

Proof. Assume the $\mathrm{secat}(p) < \infty$ and consider a corresponding minimal set of U_i's with local sections $s_i : U_i \to E$ of p. Observe that $s_i|_{B'} : U_i \cap B' \to E'$ is then a local section of p', so $\mathrm{secat}(p') \leq \mathrm{secat}(p)$. If $\mathrm{secat}(p) = \infty$, we trivially conclude $\mathrm{secat}(p') \leq \infty = \mathrm{secat}(p)$, so in either case the Schwarz genus of p' is bounded above by that of p. [1] ☐

Lemma 2.3. *For any fibration $p : E \to B$,* $\mathrm{secat}(p) \leq \mathrm{cat}(B)$.

Proof. [2] Take $\mathrm{cat}(B)$-many open sets U_i with continuous sections $s_i : U_i \to P_0B$ of $\pi_0 : P_0B \to B$. Observe that each s_i induces a homotopy $h_i : U_i \times [0, 1] \to B$ via $h_i(u, t) := s_i(u)(t)$. Then with the base point $b_0 \in B$, we can fix a corresponding element $e_0 \in p^{-1}[\{b_0\}] \subseteq E$ and define $\widetilde{h_i} : U_i \times \{0\} \to E$ by $\widetilde{h_i}(u, 0) = e_0$. This gives us that $p(\widetilde{h_i}(u, 0)) = p(e_0) = b_0 = s_i(u)(0) = h_i(u, 0)$, so by the homotopy lifting property of p there exists a homotopy $\widetilde{h_i} : U_i \times [0, 1] \to E$ such that $p \circ \widetilde{h_i} = h_i$. Then observe we can define maps $r_i : U_i \to E$ via $r_i(u) := \widetilde{h_i}(u, 1)$, giving us $p(r_i(u)) = p(\widetilde{h_i}(u, 1)) = h_i(u, 1) = u$. We therefore conclude that each r_i is a continuous section of p and so $\mathrm{secat}(p) \leq \mathrm{cat}(B)$. ☐

Proposition 2.1. *For a path-connected space X,* $\mathrm{cat}(X) \leq \mathbf{TC}_2(X) \leq \mathrm{cat}(X \times X)$.

Proof. We have the fibration $\pi : PX \to X \times X$ from before, as well as $\pi_0 : P_0X \to X \times X$ given by $\pi_0(\gamma) = (x_0, \gamma(1))$. Observe that $\mathrm{cat}(X)$ is exactly the Schwarz genus of π_0 and $P_0X = \pi^{-1}[\{x_0\} \times X]$, so we immediately know by lemma **??** that $\mathrm{cat}(X) \leq \mathbf{TC}_2(X)$. Then by lemma 2.3 we conclude $\mathbf{TC}_2(X) = \mathrm{secat}(\pi : PX \to X \times X) \leq \mathrm{cat}(X \times X)$. ☐

Theorem 2.1. $\mathbf{TC}_2$ *is a homotopy invariant.*

Definition 2.14. *The* order of instability *of the composition in definition 2.7 is the maximum r such that there is some sequence of indices $1 \leq i_1 < \cdots < i_r \leq k$ where $\overline{F_{i_1}} \cap \cdots \cap \overline{F_{i_r}}$ is non-empty. The* order of instability *of a motion planning algorithm s is then the minimum order of instability of decompositions for s.*

Definition 2.15. *The topological complexity* $\mathbf{TC}_3(X)$ *is the minimum order of instability of all tame motion planning algorithms in X.*

Remark 2.5. $\mathbf{TC}_3(X) \leq \mathbf{TC}_1(X)$.

Proof. This immediately follows from definitions 2.9 and 2.15. ☐

Definition 2.16. *A* random n-valued path *σ in X is a pair of n-tuples $(\gamma_1, ..., \gamma_n) \in (PX)^n$ and $(p_1, ..., p_n) \in [0, 1]^n$ such that all the γ_i's have the same start and end points and $p_1 + \cdots + p_n = 1$. We write σ as a formal linear combination $\sigma = p_1\gamma_1 + \cdots + p_n\gamma_n$.*

We then define P_nX to be the set of all n-valued random paths in X with the subspace topology induced by $\bigoplus_n PX$, and we have the same fibration $\pi : P_nX \to X \times X$ as before taking n-valued random paths to their end points.

[1] If one wishes to more precisely compare cardinalities, the reasoning in the finite case works.

[2] Consulted Claude for the basic idea in this proof.

Definition 2.17. *An n-*valued random motion planning algorithm *is a continuous section $s : X \times X \to P_n X$ of π.*

Observe that $s(x, y)$ is simply a probability distribution on n paths from x to y.

Definition 2.18. *The topological complexity $\mathbf{TC}_4(X)$ is the minimum integer n such that there is some n-valued random motion planning algorithm $s : X \times X \to P_n X$.*

Theorem 2.2. *If X is a simplicial polyhedron, then*

$$\mathbf{TC}_1(X) = \mathbf{TC}_2(X) = \mathbf{TC}_3(X) = \mathbf{TC}_4(X).$$

Proof sketch. The proof proceeds in steps:

Step 1. $\mathbf{TC}_1(X) \leq \mathbf{TC}_2(X)$: Expand each F_i to a U_i that deformation retracts back onto it and use the deformation retract to extend $s|_{F_i} : F_i \to PX$ so $s_i : U_i \to PX$.

Step 2. $\mathbf{TC}_2(X) \leq \mathbf{TC}_1(X)$: Trim each U_i just enough to get disjoint F_i's that are ENRs.

Step 3. $\mathbf{TC}_3(X) \leq \mathbf{TC}_2(X)$: Same basic idea as step 1.

Step 4. $\mathbf{TC}_1(X) \leq \mathbf{TC}_3(X)$: Done already in remark 2.5.

Step 5. $\mathbf{TC}_2(X) \leq \mathbf{TC}_4(X)$: Construct open sets $U_j \subseteq X \times X$ from the functions $p_j : X \times X \to [0, 1]$, and conclude by construction.

Step 6. $\mathbf{TC}_4(X) \leq \mathbf{TC}_2(X)$: Extend the s_i's arbitrarily, weight them via a partition of unity, and conclude.

□

Definition 2.19. *Moving forward, we define the* topological complexity *of a space X to be $\mathbf{TC}(X) := \mathbf{TC}_2(X)$.*

Definition 2.20. *The* order *of an open cover $\mathcal{U}$ of X is the least integer n such that each point of X belongs to at most n sets of $\mathcal{U}$ (if such an integer exists) [Wik26d].*

Definition 2.21. *The* covering dimension *of a space X, denoted $\dim X$, to be the least integer n such that every finite open cover $\mathcal{U}$ of X has a refinement $\mathcal{V}$ of order $n + 1$. If no such n exists, then we set $\dim X = \infty$ [Wik26d].*

Lemma 2.4. *Let X be a path-connected paracompact locally contractible space. Then*

$$\operatorname{cat}(X) \leq \dim(X) + 1.$$

Lemma 2.5. *Let X be a path-connected paracompact locally contractible space. Then*

$$\dim(X \times X) = 2 \dim X.$$

Theorem 2.3. *Let X be a path-connected paracompact locally contractible space. Then*

$$\mathbf{TC}(X) \leq 2 \dim X + 1.$$

Proof. By lemmas 2.3, 2.4, and 2.5 we have

$$\mathbf{TC}(X) \leq \mathrm{cat}(X \times X) \leq \dim(X \times X) + 1 = 2\dim X + 1.$$

□

Theorem 2.4. *Let X be an r-connected CW-complex. Then*

$$\mathbf{TC}(X) \leq \frac{2\dim X + 1}{r+1} + 1.$$

Definition 2.22. *The tensor product $H^\bullet(X) \otimes H^\bullet(X)$ is a graded algebra with multiplication given by*

$$(u_1 \otimes v_1) \cdot (u_2 \otimes v_2) = (-1)^{|v_1||u_2|}(u_1 u_2) \otimes (v_1 v_2)$$

where $|x|$ denotes the degree of the cohomology class of x.

Definition 2.23. *The* cup product $\smile : H^k(X) \otimes H^\ell(X) \to H^{k+\ell}(X)$ *is given by*

$$(\alpha \smile \beta)(\sigma) := \alpha(\sigma_{[0,\dots,k]}) \cdot \beta(\sigma_{[k,\dots,k+\ell]}).$$

Note that this means $\smile$ induces a graded algebra $H^\bullet(X)$.

Definition 2.24. *The* ideal of the zero-divisors *of $H^\bullet(X)$ is the kernel of*

$$\smile : H^\bullet(X) \otimes H^\bullet(X) \to H^\bullet(X).$$

Definition 2.25. *The* zero-divisors-cup-length *or simply* cup-length *of $H^\bullet(X)$, denoted* $\mathrm{cup}(X)$ *is the length of the longest nontrivial product in the ideal of zero-divisors of $H^\bullet(X)$.*

Theorem 2.5. $\mathbf{TC}(X)$ *is greater than the zero-divisors-cup-length of $H^\bullet(X)$.*

Example 2.1. *Consider S^n, letting $u \in H^n(S^n)$ be the fundamental class and $1 \in H^0(S^n)$ be the unit. Recall that $H^k(S^n) = 0$ for $k \neq 0, n$, and note $H^n(S^n) = \langle u \rangle$ and $H^0(S^n) = \langle 1 \rangle$. We then set $a = u \otimes 1 - 1 \otimes u$ and $b = u \otimes u$, and note that, up to coefficients, a and b are the only (potentially) interesting elements of $H^\bullet(S^n) \otimes H^\bullet(S^n)$. Observe that $u^2 \in H^{2n}(S^n) = 0$ so $u^2 = 0$ and we compute*

$$\begin{aligned}
a \cdot a &= (u \otimes 1 - 1 \otimes u) \cdot (u \otimes 1 - 1 \otimes u) \\
&= (u \otimes 1) \cdot (u \otimes 1) - (u \otimes 1) \cdot (1 \otimes u) - (1 \otimes u) \cdot (u \otimes 1) + (1 \otimes u) \cdot (1 \otimes u) \\
&= (-1)^{|1||u|}(u^2 \otimes 1^2) - (-1)^{|1||1|}(u \otimes u) - (-1)^{|u||u|}(u \otimes u) + (-1)^{|u||1|}(1^2 \otimes u^2) \\
&= -u \otimes u - (-1)^{n^2}(u \otimes u) \\
&= ((-1)^{n+1} - 1) u \otimes u.
\end{aligned}$$

So if n is even then $a^2 = -2b$ and if n is odd $a^2 = 0$, giving us that the zero-divisors-cup-length of $H^\bullet(S^n)$ is 1 for odd n and 2 for even n.

Theorem 2.6. *For spaces X and Y,*

$$\mathbf{TC}(X \times Y) \leq \mathbf{TC}(X) + \mathbf{TC}(Y) - 1.$$

Corollary 2.1. *For spaces $X_1, ..., X_n$ with $\mathbf{TC}(X_i) \leq M$ for all $1 \leq i \leq n$,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n) \leq nM - 1.$$

Theorem 2.7. *For n path-connected spaces X_i that are (co)homologically nontrivial,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n) \geq n + 1.$$

Corollary 2.2. *For n path-connected spaces X_i that are (co)homologically nontrivial,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n)$$

grows linearly with n.

Proof. This follows directly from corollary 2.1 and theorem 2.7. ☐

Proposition 2.2. *For $U \subseteq X \times X$, there exists a section $s : U \to PX$ of $\pi : PX \to X \times X$ if and only if the inclusion $i : U \hookrightarrow X \times X$ is homotopic to some diagonal map $f : U \to X \times X$ (i.e. $f[U]$ is contained in the diagonal of $X \times X$).*

Proof. First assume we have a section $s : U \to PX$ of π and consider the diagonal map $f : U \to X \times X$ given by $f(x, y) := (y, y)$. Then we can construct a homotopy $h_t : U \to X \times X$ by $h_t(x, y) := (s(x, y)(t), y)$. Observe that $h_0(x, y) = (x, y) = i(x, y)$ and $h_1(x, y) = (y, y) = f(x, y)$ so we have a homotopy from $i : U \to X \times X$ to a diagonal map $f : U \to X \times X$.

Conversely, assume we have a homotopy $h_t : U \to X \times X$ such that $h_0 = i$ and $h_1 =: f$ is a diagonal map; denote $\widehat{f} = f_1 = f_2$. Then for some $(x, y) \in U$, we have two paths, $\gamma_1(t) = (h_t(x, y))_1$ from x to $\widehat{f}(x, y)$ and $\gamma_2(t) = (h_t(x, y))_2$ from y to $\widehat{f}(x, y)$. Then we can define $s : U \to PX$ by $s(x, y) = \overline{\gamma_2} * \gamma_1$, and by construction we have $\pi \circ s = \mathrm{id}$ ☐

Corollary 2.3. *The topological complexity $\mathbf{TC}(X) = \mathbf{TC}_2(X)$ is the smallest integer k such that $X \times X$ can be covered by k open sets where each inclusion $U_j \hookrightarrow X \times X$ is homotopic to a diagonal map $f : U \to X \times X$.*

Proof. This follows immediately from definition 2.13 and proposition 2.2. ☐

Proposition 2.3. *If X is an ENR then*

$$\mathbf{TC}(X) \geq \mathrm{cat}((X \times X)/\Delta X) - 1$$

where $\Delta X \subseteq X \times X$ is the diagonal.

2.1.2 Background from [CG13]

Definition 2.26. *The* sectional category *of a map* $p : E \to B$*, denoted* $\mathrm{secat}(p)$*, is the least integer* k *such that* B *may be covered by* k *open sets* U_j *that each admit a map* $s_j : U_j \to E$ *where* $p \circ s_j$ *is homotopic to the inclusion* $i_j : U_j \hookrightarrow B$*.*

Remark 2.6. *For a fibration* $p : E \to B$*, definition 2.26 corresponds to definition 2.10 given in [Far06].*

Proof. First, let U_j and $s_j : U_j \to E$ be as in definition 2.10, i.e. $p \circ s_j = i_j$. Then trivially $p \circ s_i \simeq i_j$.

Conversely, let U_j and $s_j : U_j \to E$ be as in definition 2.26, i.e. $p \circ s_j \simeq i_j$, and fix some $j \leq k$. Let $h_t : U_j \to B$ be the homotopy witnessing $p \circ s_j \simeq i_j$ with $h_0 = p \circ s_j$ and $h_1 = i_j$. We define the map $\widetilde{h_0} : U_j \to E$ by $\widetilde{h_0} = s_j$, and observe that by construction $p \circ \widetilde{h_0} = p \circ s_j = h_0$. Then by the homotopy lifting property we have a homotopy $\widetilde{h_t} : U_j \to E$ such that $p \circ \widetilde{h_t} = h_t$. In particular, this gives us a map $\widehat{s_j} := \widetilde{h_1} U_j \to E$ such that $p \circ \widehat{s_j} = h_1 = i_j$. □

Definition 2.27. *The* Lusternik-Schnirelmann category *of a space* X*, denoted* $\mathrm{cat}(X)$*, is the least integer* k *such that* X *may be covered by* k *open sets* U_j *where each inclusion* $i_j : U_j \hookrightarrow X$ *is null-homotopic.*

Remark 2.7. *For a path-connected space* X*, definition 2.27 corresponds to definition 2.12 given in [Far06]. Note that definition 2.12 only makes sense if* X *is path-connected anyway.*

Proof. First, let $U_j \subseteq X$ and $s_j : U_j \to P_0X$ be as in definition 2.12, i.e. $\pi_0 \circ s_j = i_j$. We then simply define a homotopy $h_t : U_j \to X$ via $h_t(x) = s_j(x)(t)$, so $h_0(x) = s_j(x)(0) = x_0$ and $h_1(x) = s_j(x)(1) = x$. Therefore $h_0 = c_{x_0}$ and $h_1 = i_j$, so $c_{x_0} \simeq i_j$.

Conversely, let $U_j \subseteq X$ be as in definition 2.27 with $i_j \simeq c_{x_j}$ for some $x_j \in X$. Since X is path-connected all constant maps are homotopic, so without loss of generality each $x_j = x_0$. Fixing some j, let $h_t : U_j \to X$ be the homotopy witnessing $i_j \simeq c_{x_0}$ where $h_0 = c_{x_0}$ and $h_1 = i_j$. Then we define $s_j : U_j \to P_0X$ by $s_j(x)(t) = h_t(x)$, so

$$\pi_0(s_j(x)) = s_j(x)(1) = h_1(x) = i_j(x) = x.$$

□

Proposition 2.4. *For any spaces* E *and* B *along with a surjective map* $p : E \to B$*,* $\mathrm{secat}(p) \leq \mathrm{cat}(B)$*.*

Proof. Take $\mathrm{cat}(B)$-many open sets U_j such that $B = \cup_j U_j$ and each inclusion $i_j : U_j \hookrightarrow B$ is homotopic to a constant map $c_j : U_j \to B$. For each j, fix $e_j \in p^{-1}[c_j[U_j]]$ and define $s_j : U_j \to E$ by $s_j = c_{e_j}$. Observe that $p \circ s_j = c_j$ by definition, so $p \circ s_j \simeq i_j$. Therefore $\mathrm{secat}(p) \leq \mathrm{cat}(B)$. □

Proposition 2.5. *For a contractible space* E *and any space* B *along with a map* $p : E \to B$*,* $\mathrm{cat}(B) \leq \mathrm{secat}(p)$*.*

Proof. Take $\operatorname{secat}(p)$-many open sets U_j with $s_j : U_j \to E$ such that $B = \cup_j U_j$ and $p \circ s_j \simeq i_j$. Since E is contractible we have that each map s_j is homotopic to a constant map c_j, and immediately this means that $p \circ s_j$ is homotopic to $p \circ c_j$ which is also constant. Therefore $i_j \simeq p \circ c_j$ and U_j is null-homotopic so $\operatorname{cat}(B) \leq \operatorname{secat}(p)$. □

Corollary 2.4. *If E is a contractible space, B is any space, and $p : E \to B$ is a surjection, then* $\operatorname{secat}(p) = \operatorname{cat}(B)$.

Proof. This directly follows from propositions 2.4 and 2.5. □

Definition 2.28. *Given a commutative ring R and an ideal $I \subseteq R$, the* nilpotency *of I, denoted* $\operatorname{nil} I$, *is the maximum integer k such that we can find elements $r_1, ..., r_k \in I$ with $r_1 \cdots r_k \neq 0$.*

Proposition 2.6. *Let $p : E \to B$ be a fibration and let $p^* : H^*(B) \to H^*(E)$ be the induced homomorphism on cohomology. Then* $\operatorname{secat}(p) > \operatorname{nil} \ker p^*$.

Definition 2.29. *Let G be a topological group and X be a space, and assume we have some continuous G-action on X (i.e. $(g, x) \mapsto g \cdot x$ is continuous). Then X is a G-*space.

Definition 2.30. *For $x \in X$ a G-space, the* isotropy group *or* stabilizer group *of x is*

$$G_x := \{g \in G : gx = x\}.$$

Remark 2.8. *G_x is closed for all $x \in X$.*

Proof. Fix some $x \in X$ and consider a point $g \in \overline{G_x}$. Then there is a net $\{g_d\}_{d \in D} \subseteq G_x$ such that $g_d \to g$. Since each $g_d x = x$ by definition, by continuity we have that $g_d x \to gx = x$, so $g \in G_x$ and G_x is closed. □

Definition 2.31. *Given a G-space X and $x \in X$, the* orbit *of x is*

$$Gx := \{gx : g \in G\}.$$

Remark 2.9. *The map $\phi : G/G_x \to Gx$ given by $\phi(gG_x) = gx$ is a homeomorphism.*

Proof. First, let $g, g' \in G$ such that $gG_x = g'G_x$; we wish to show that $gx = g'x$. Since $gG_x = g'G_x$, there exists some $h \in G_x$ such that $ge = g'h$. Then we have

$$gx = gex = g'hx = g'x.$$

So ϕ is well-defined.

To see that ϕ is continuous, consider the quotient map $q : G \to G/G_x$; we wish to show that $\phi \circ q : G \to Gx$ is continuous. Observe that $\phi(q(g)) = \phi(gG_x) = gx$, so it immediately follows that $\phi \circ q$ is continuous.

Lastly, to see that ϕ is an open map, consider some open set $UG_x \subseteq G/G_x$. Then $\phi[UG_x] = Ux$ which is clearly open.

Therefore we conclude that ϕ is a homeomorphism. □

Definition 2.32. *For a G-space X, the* orbit space *is X/G.*

Proposition 2.7. *If G is compact and X is a Hausdorff G-space, then X/G is Hausdorff.*

Proof. [3] First consider the equivalence relation given on X where xRy if x and y lie in the same orbit. This equivalence relation is then given by the following set:

$$R := \{(x, gx) : x \in X, g \in G\} = \bigcup_{x \in X} \{x\} \times Gx \subseteq X \times X.$$

We wish to show that R is closed, so let $\{(x_a, g_a x_a)\}_{a \in A} \subseteq R$ be a convergent net with $(x_a, g_a x_a) \to (x, y) \in X \times X$. Since G is compact, the net $\{g_a\}_{a \in A}$ has a convergent subnet $\{g_b\}_{b \in B}$ with $g_b \to g \in G$. Then by continuity we have that the subnet $\{(x_b, g_b x_b)\}_{b \in B}$ converges to $(x, gx) \in R$ so $y = gx$. Thus R is closed.

Additionally, observe that $x \mapsto gx$ is a homeomorphism for any $g \in G$, so for any open $U \subseteq X$, $GU = \cup_{g \in G} gU$ is open as well. Then with the quotient map $q : X \to X/G$, $q^{-1}[GU] = GU$ since GU is a union of orbits, so $GU \in X/G$ is open.

Now let $Gx, Gy \in X/G$ with $Gx \neq Gy$, so clearly x and y do not lie in the same orbit and $(x, y) \notin R$. Because R is closed in $X \times X$, we can find open sets $U, V \subseteq X$ such that $(x, y) \in U \times V$ and $(U \times V) \cap D = \varnothing$. We immediately have that $GU \cap GV = \varnothing$ as well since otherwise x and y would be in the same orbit. Therefore we conclude that X/G is Hausdorff. $\square$

Definition 2.33. *Given two G-spaces X and Y, $f : X \to Y$ is a G-*map *if for all $g \in G$, $f(gx) = g \cdot f(x)$. We write $f : X \to_G Y$.*

Definition 2.34. *Given two G-spaces X and Y, $h : X \times [0, 1] \to Y$ is a G-*homotopy *if h is a homotopy and $h(-, t)$ is a G-map for all $t \in [0, 1]$. Equivalently, h is a G-map with G acting trivially on $[0, 1]$ and diagonally on $X \times [0, 1]$. We write $h : X \times [0, 1] \to_G Y$ or $h_t : X \to_G Y$.*

Definition 2.35. *Two G-maps $f, g : X \to_G Y$ are G-*homotopic *if there is a G-homotopy $h_t : X \to_G Y$ such that $h_0 = f$ and $h_1 = g$. We write $f \simeq_G g$.*

Definition 2.36. *Two G-spaces X and Y are G-*homotopy equivalent *if there are G-maps $f : X \to_G Y$ and $g : Y \to_G X$ such that $g \circ f \simeq_G \mathrm{id}_X$ and $f \circ g \simeq_G \mathrm{id}_Y$.*

Definition 2.37. *A set $U \subseteq X$ is* invariant *if $gU \subseteq U$ for all $g \in G$. Note that $U \subseteq GU$ always, so U is invariant if and only if $U = GU$.*

Definition 2.38. *A G-map $f : X \to Y$ is a* fixed orbit map *if $f[X]$ is contained in a single orbit in Y.*

Definition 2.39. *An invariant set $U \subseteq X$ is G-*categorical *if the inclusion $i : U \to X$ is G-homotopic to a fixed orbit map.*

Definition 2.40. *The* equivariant category *of a G-space X, denoted $\mathrm{cat}_G(X)$, is the least integer k such that X can be covered by k G-categorical open sets.*

Definition 2.41. *A G-space X is G-*contractible *if $\mathrm{cat}_G(X) = 1$.*

[3] Got proof idea from this conversation with Claude.

Proposition 2.8. *X is G-contractible if and only if X is G-homotopy equivalent to G.*

Proof. First assume that X is G-contractible, and let $h_t : X \to_G X$ be a corresponding G-homotopy witnessing this where $h_0 = \mathrm{id}_X$ and h_1 is a fixed orbit map. Let $x_0 \in X$ be some representative of the orbit $h_1[X]$, i.e. $Gx_0 = h_1[X]$. Observe that Gx_0 is trivially homeomorphic to G via the map $gx_0 \mapsto g$, so we conclude that $X \simeq_G G$.

Conversely, assume that $X \simeq_G G$ via $f : X \to_G G$ and $g : G \to_G X$ with corresponding G-homotopy $h_t : X \to_G X$ where $h_0 = g \circ f$ and $h_1 = \mathrm{id}_X$. Observe that g must be a fixed orbit map since $g(h \cdot e) = h \cdot g(e)$ so $g[G] = Gg(e)$. Then $g \circ f$ is trivially a fixed orbit map since $(g \circ f)[X] \subseteq g[G]$, so we conclude that X is G-contractible via h_t. □

Corollary 2.5. *For any topological group G acting on itself in the usual way, we have that* $\mathrm{cat}_G(G) = 1$.

Proof. Clearly $G \simeq_G G$, so by proposition 2.8 we are done. □

Proposition 2.9. *For any G-space X,* $\mathrm{cat}_G(X) \geq \mathrm{cat}(X/G)$.

Proof. Take $\mathrm{cat}_G(X)$-many G-categorical open sets U_j that cover X. Since each U_j is invariant, $U_j = GU_j \subseteq X/G$ is open. Then $i_j : U_j \hookrightarrow X$ is G-homotopic to a fixed orbit map, so immediately we have that $q \circ i_j : GU_j \hookrightarrow X/G$ is homotopic to a constant map and each GU_j is null homotopic with $X/G = \cup_j GU_j$. □

Proposition 2.10. *If X is metrizable and acted freely upon by G, then* $\mathrm{cat}_G(X) = \mathrm{cat}(X/G)$.

Definition 2.42. *The* equivariant sectional category *of a G-map $p : E \to_G B$, denoted* $\mathrm{secat}_G(p)$, *is the least integer k such that B can be covered by k invariant open sets U_j that each admit a local G-homotopy section, i.e. there exists $s_j : U_j \to E$ such that $p \circ s \simeq_G i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion.*

Proposition 2.11. *If $p : E \to_G B$ is a G-fibration, then* $\mathrm{secat}_G(p) \leq k$ *if and only if B can by covered by k invariant open sets that each admit a local G-section.*

Proof. This follows by an argument analogous to remark 2.6. □

Definition 2.43. *Given a G-space X and a closed subgroup $H \subseteq G$, the H-*fixed point set *of X is*

$$X^H := \{x \in X : \forall h \in H, hx = x\} = \{x \in X : Hx = \{x\}\}.$$

Definition 2.44. *A G-space X is G-*connected *if X^H is path-connected for every closed subgroup $H \subseteq G$.*

Proposition 2.12. *Let $p : E \to B$ be a G-map where B is G-connected and $E^G \neq \varnothing$. Then* $\mathrm{secat}_G(p) \leq \mathrm{cat}_G(B)$.

Proposition 2.13. *For a G-map $p : E \to_G B$, if $p[E^H] = B^H$ for all closed subgroups $H \subseteq G$, then* $\mathrm{secat}_G(p) \leq \mathrm{cat}_G(B)$.

Proposition 2.14. *If E is G-contractible then for any G-map $p : E \to_G B$ we have* $\mathrm{cat}_G(B) \leq \mathrm{secat}_G(p)$.

Proof. Since E is G-contractible, we have that $\mathrm{id}_E \simeq_G c$ where $c : E \to_G E$ is a fixed orbit map. Now take $\mathrm{secat}_G(p)$-many invariant open sets $U_j \subseteq B$ such that $B = \cup_j U_j$ with maps $s_j : U_j \to E$ such that $s_j \simeq_G i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion. Then we have

$$i_j \simeq_G p \circ s_j = p \circ \mathrm{id}_E \circ s_j \simeq_G p \circ c \circ s_j$$

in which the latter map is a fixed orbit map since c is a fixed orbit map. Hence we conclude $\mathrm{cat}_G(B) \leq \mathrm{secat}_G(p)$. For clarity, this is represented in the following commutative diagram:

$$\begin{tikzcd}
E \arrow[r, shift left, "\mathrm{id}_E"] \arrow[r, shift right, "c"'] \arrow[dr, "p"'] & E \arrow[d, "p"] \\
U_j \arrow[u, "s_j"] \arrow[r, "i_j"'] & B
\end{tikzcd}$$

□

Corollary 2.6. *Let $p : E \to_G B$ be a G-map and let E be a G-contracible space such that one of the following holds:*

1. *B is G-connected and $E^G \neq \varnothing$*
2. *$p[E^H] = B^H$ for all closed subgroups $H \subseteq G$.*

Then $\mathrm{secat}_G(p) = \mathrm{cat}_G(p)$.

Proof. This immediately follows from propositions 2.12, 2.13, and 2.14. □

2.1.3 Background from [DLS24]

Definition 2.45. *A map $f : X \to Y$ is* $(G\text{-})$invariant *if for all $x \in X$ and for all $g \in G$, $f(g \cdot v) = f(v)$.*

Definition 2.46. *For a space X and group G, we denote X as a G-space with G acting trivially by X_{triv}.*

Definition 2.47. *We denote the category of topological spaces as* ***Top*** *and the category of G-spaces for fixed group G as G-**Top**.*

Remark 2.10. *A map $f : X \to Y$ is invariant if and only if it is a G-map when G acts trivially on Y, i.e. $f : X \to_G Y_{triv}$.*

Remark 2.11. *An invariant map $f : X \to Y$ induces a map $\widehat{f} : X/G \to Y$ via $\widehat{f}(Gv) = f(v)$. Conversely, a map $\widehat{f} : X/G \to Y$ induces an invariant map $f : X \to Y$ via $f(v) = \widehat{f}(Gv)$. This gives an isomorphism $\boldsymbol{Top}(X/G, Y) \cong G\text{-}\boldsymbol{Top}(X, Y_{triv})$.*

Furthermore, let $L : G\text{-}\boldsymbol{Top} \to \boldsymbol{Top}$ and $R : \boldsymbol{Top} \to G\text{-}\boldsymbol{Top}$ be the functors defined as follows:

- *$LX = X/G$, $L(f : X \to_G Y) = (Gx \mapsto G(f(x)))$*
- *$RY = Y_{triv}$, $R(f : Y \to X) = (y \mapsto f(y))$.*

Then the previous isomorphism actually means that L is a left adjoint of R.

Definition 2.48. *Given a G-space X, an* orbit canonicalization *is an invariant map $f : X \to X$ such that $f(x) \in Gx$ for all $x \in X$.*

Remark 2.12. *An orbit canonicalization $f : X \to X$ must have a unique fixed point in each orbit $Gx \in X/G$.*

Proof. Remark 2.11 gives us an embedding $X/G \hookrightarrow X$, and we conclude by instead considering this embedding as a map $X \to X$. $\square$

Proposition 2.15. *Every orbit canonicalization $f : X \to X$ corresponds uniquely to a continuous section $s : X/G \to X$.*

Proof. First, let $f : X \to X$ be an orbit canonicalization. As per remark 2.11, f induces $\widehat{f} : X/G \to X$ via $\widehat{f}(Gx) = f(x)$ with $\widehat{f} \circ q = f$. Since f is invariant, we have that $q(\widehat{f}(Gx)) = q(f(x)) = G(f(x)) = Gx$ and so $\widehat{f}$ is a section of q.

Conversely, let $s : X/G \to X$ be some section of q such that $s(Gx) \in Gx$ always. Then we can define a map $f : X \to X$ via $f = s \circ q$ and observe that $f(x) = s(q(x)) = s(Gx)$. Since s is a section of q, we have that $q(s(Gx)) = Gx$, meaning that $f(x) = s(Gx) \in q^{-1}[Gx] = Gx$. Therefore f is an orbit canonicalization.

Lastly, to see the uniqueness, observe that the process in the converse step applied to $\widehat{f}$ yields f, so the map $f \mapsto \widehat{f}$ is a bijection. $\square$

Corollary 2.7. *An orbit canonicalization $f : X \to X$ exists if and only if there is a continuous section $s : X/G \to X$.*

Proof. This is just a restatement of proposition 2.15. $\square$

2.1.4 Stiefel-Whitney Classes

The following material is from [Hat09]

Definition 2.49. *Let E and B be spaces and let $p : E \to B$ be a map such that*

1. *p is surjective*
2. *for all $b \in B$, $p^{-1}[\{b\}]$ has an $\mathbb{R}$-vector space structure,*
3. *and there is a cover of B by open sets U_α with homeomorphisms $\Phi_\alpha : p^{-1}[U_\alpha] \to U_\alpha \times \mathbb{R}^n$ such that $\Phi_\alpha|_{p^{-1}[\{b\}]} : p^{-1}[\{b\}] \to \{b\} \times \mathbb{R}^n$ is a linear isomorphism for each $b \in U_\alpha$.*

*Then p is an n-*dimensional vector bundle *and the maps Φ_α are* local trivializations*.*

Remark 2.13. *An n-dimensional vector bundle is a special type of fiber bundle with fiber $\mathbb{R}^n$.*

Proposition 2.16. *A vector bundle $p : E \to B$ is trivial (i.e. isomorphic to the product $B \times \mathbb{R}^n$) if and only there are n linearly independent sections $s_1, ..., s_k : B \to E$ of p.*

Definition 2.50. *Let $p : E \to B$ be a vector bundle and let $f : B' \to B$ be some map. Then we can define the* pullback bundle *induced by f to be*

$$E' = f^*(E) := \{(b', e) \in B \times E : f(b') = p(e)\}.$$

Remark 2.14. *From definition 2.50, the map $p' : E' \to B'$ given by $p'(b', e) = b'$ is also a vector bundle. Additionally, E' is the unique pullback $B' \times_B E$ in the category* ***Top****.*

Definition 2.51. *The* Grassmann manifold $\mathrm{Gr}(n, k)$ *is the space of all k-dimensional vector subspaces of $\mathbb{R}^n$.*

Remark 2.15. *There is a natural vector bundle over $\mathrm{Gr}(n, k)$ given by*

$$E(n, k) := \{(P, x) \in \mathrm{Gr}(n, k) \times \mathbb{R}^n : x \in P\}$$

with the projection $p : E(n, k) \to \mathrm{Gr}(n, k)$.

Definition 2.52. *The* Stiefel manifold $\mathrm{St}(n, k)$ *is the space of all orthonormal bases for k-dimensional vector subspaces of $\mathbb{R}^n$.*

Remark 2.16. *There is a natural vector bundle over $\mathrm{St}(n, k)$ given by*

$$F(n, k) := \{(B, x) \in \mathrm{St}(n, k) \times \mathbb{R}^n : x \in \operatorname{span} B\}$$

with the projection $q : F(n, k) \to \mathrm{Gr}(n, k)$. This bundle is trivial by proposition 2.16 since one clearly has k orthogonal (and hence linearly independent) sections of q given by $s_i : \mathrm{St}(n, k) \to F(n, k)$, $s_i(B) = (B, b_i)$ where we write B in the standard basis of $\mathbb{R}^n$ with $B = [b_1 \cdots b_k]$.

Remark 2.17. *There is a natural projection $\pi : \mathrm{St}(n, k) \to \mathrm{Gr}(n, k)$ given by $\pi(B) = \operatorname{span} B$.*

Theorem 2.8. *(Theorem 3.1 in [Hat09]) There exist unique functions, the* Stiefel-Whitney classes*, $w_1, w_2, ...$ that map each real vector bundle $E \to B$ to a class $w_i(E) \in H^i(B; \mathbb{Z}_2)$ such that*

1. *If $E \cong E'$ as vector bundles, $w_i(E) = w_i(E')$.*
2. *For a function $f : A \to B$ with induced map on homology $f^* : H^*(B; \mathbb{Z}_2) \to H^*(A; \mathbb{Z}_2)$ and pullback $f^*(E)$, we have $w_i(f^*(E)) = f^*(w_i(E))$.*
3. *$w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$ where $w = 1 + w_1 + w_2 + \cdots \in H^*(B; \mathbb{Z}_2)$.*
4. *$w_i(E) = 0$ if $i > \dim E$.*
5. *For the canonical line bundle $E \to \mathbb{R}P^\infty$, $w_1(E)$ is a generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2)$.*

Remark 2.18. *A trivial vector bundle has $w_i(E) = 0$ for $i > 0$.*

Proposition 2.17. *(From [Jaw06]) Let $w_1, ..., w_k$ be the Stiefel-Whitney classes of the bundle $p : E(n, k) \to \mathrm{Gr}(n, k)$, let $\overline{w_1}, ..., \overline{w_n}$ be the dual classes, and let $J(n, k)$ be the ideal in $\mathbb{Z}_2[w_1, ..., w_k]$ generated by the relation $w\overline{w} = 1$. Then*

$$H^*(\mathrm{Gr}(n, k)) \cong \mathbb{Z}_2[w_1, ..., w_k]/J(n, k).$$

Proposition 2.18. *Let $\pi : \mathrm{St}(n, k) \to \mathrm{Gr}(n, k)$ be the natural projection. Then the induced map $\pi^* : H^*(\mathrm{Gr}(n, k)) \to H^*(\mathrm{St}(n, k))$ is zero on positive-dimensional cohomology.*

Proof. Let $\pi^*(E(n, k))$ be the pullback of the vector bundle $p : E(n, k) \to \mathrm{Gr}(n, k)$ via $\pi : \mathrm{St}(n, k) \to \mathrm{Gr}(n, k)$. Then observe that

$$\begin{aligned}
\pi^*(E(n, k)) &= \{(B, (P, x)) \in \mathrm{St}(n, k) \times E(n, k) : \pi(B) = p(P, x) = P\} \\
&= \{(B, x) \in \mathrm{St}(n, k) \times \mathbb{R}^n : x \in \operatorname{span} B\} \\
&= F(n, k).
\end{aligned}$$

Thus the pullback bundle is actually just the bundle $q : F(n, k) \to \mathrm{St}(n, k)$. Then by theorem 2.8 we have that $\pi^*(w_i(E(n, k))) = w_i(F(n, k)) = 0$ since $q : F(n, k) \to \mathrm{St}(n, k)$ is a trivial bundle. By proposition 2.17 we know that $H^*(\mathrm{Gr}(n, k); \mathbb{Z}_2)$ is generated by (a quotient of) the Stiefel-Whitney classes $w_i(E(n, k))$, so we conclude that $\pi^* = 0$ on positive-dimensional cohomology. ☐

Corollary 2.8. $\operatorname{nil}\ker \pi^* = \operatorname{cup}(\mathrm{Gr}(n, k)) - 1$

2.2 New Work

2.2.1 Sectional Category and Canonicalization

Proposition 2.19. *If there exists an orbit canonicalization $f : X \to X$, then* $\mathrm{secat}(q) = 1$ *where $q : X \to X/G$ is the quotient map.*

Proof. By proposition 2.15 we know that f induces a section $\widehat{f} : X/G \to X$ of q, so we conclude that $\mathrm{secat}(q) = 1$. $\square$

Corollary 2.9. *If* $\mathrm{secat}(q) > 1$*, there is no orbit canonicalization $f : X \to X$.*

With this in mind, we now wish to study the sectional category of the quotient map $q : X \to X/G$.

Definition 2.53. *For spaces E and B, a map $p : E \to B$ has the* homotopy lifting property up to homotopy (HLPH) *for a space X if for every homotopy $h_t : X \to B$ and every homotopy lift $\widetilde{h}_0 : X \to E$ of h_0 ($h_0 \simeq p \circ \widetilde{h}_0$), there exists a homotopy $\widetilde{h}_t : X \to E$ that lifts h_t up to homotopy ($h_t \simeq p \circ \widetilde{h}_t$).*

Proposition 2.20. *For any spaces E and B and a surjective map $p : E \to B$,* $\mathrm{secat}(p) \leq \mathrm{cat}(B)$ *as in definitions 2.10 and 2.12.*

Proof. Take $\mathrm{cat}(B)$-many open sets U_j such that $B = \cup_j U_j$ and each inclusion $i_j : U_j \hookrightarrow B$ is homotopic to a constant map $c_j : U_j \to B$. $\square$

Definition 2.54. *For spaces E and B, a map $p : E \to B$ is a* fibration up to homotopy *if p satisfies the HLPH for all topological spaces.*

Remark 2.19. *Clearly HLP implies $HLPH$ and any fibration is a fibration up to homotopy.*

Lemma 2.6. *Let X be a G-space and let $q : X \to X/G$ be the quotient map. Then if $U \subseteq X$ is G-categorical, the set $q[U] \subseteq X/G$ is contractible.*

Proof. First, let $U \subseteq X$ be G-categorical, i.e. the inclusion $i : U \hookrightarrow X$ is G-homotopic to a fixed orbit map $c : U \to X$. This clearly induces a homotopy from the inclusion $q \circ i : q[U] \hookrightarrow X$ to $q \circ c : q[U] \to X/G$, and since c is a fixed orbit map $q \circ c$ must be a constant map. Thus $q[U]$ is contractible. $\square$

Lemma 2.7. *Let X be a G space such that the quotient map $q : X \to X/G$ is a fibration. Then if $U \subseteq X/G$ is open and contractible, the set $q^{-1}[U] \subseteq X$ is G-categorical.*

Proof. Let $h_t : U \to X/G$ be a homotopy witnessing that U is contractible, with h_0 being the inclusion $i : U \hookrightarrow X/G$ and h_1 being a constant map $c : U \to X/G$. Set $V = q^{-1}[U]$, noting that V is open and invariant, and let $\widetilde{h}_0 : V \to X$ be the inclusion $\widetilde{i} : V \hookrightarrow X$. Then clearly $q \circ \widetilde{h}_0 = i$, so by the homotopy lifting property we have a homotopy $\widetilde{h}_t : V \to X$ such that $q \circ \widetilde{h}_t = h_t$. Specifically this means $q \circ \widetilde{h}_1 = c$, i.e. $\widetilde{h}_1 : V \to X$ is a fixed orbit map, so we conclude that V is G-categorical. $\square$

Proposition 2.21. *If X is a G-space such that the quotient map $q : X \to X/G$ is a fibration up to homotopy, then* $\mathrm{secat}(q) \leq \mathrm{cat}_G(X)$*.*

Proof. Take $\mathrm{cat}_G(X)$-many G-categorical sets $U_j \subseteq X$ such that $X = \cup_j U_j$. Define $V_j = q[U_j]$, and note that by lemma 2.6 each V_j is contractible. Fix some j and let $h_t : V_j \to X/G$ be a homotopy witnessing the contractibility of V_j, with h_0 being a constant map $c_j : V_j \to X/G$ and h_1 being the inclusion $i_j : V_j \hookrightarrow X/G$. Then fix some $x_0 \in q^{-1}[c_j[V_j]]$ and define $\widetilde{h}_0 : V_j \to X$ by $\widetilde{h}_0 = c_{x_0}$. Clearly $q \circ \widetilde{h}_0 = c_j$, so the HLPH gives us a homotopy $\widetilde{h}_t : V_j \to X$ such that $q \circ \widetilde{h}_t \simeq h_t$. In particular, this means $q \circ \widetilde{h}_1 \simeq i_j$ so $\widetilde{h}_1$ is a homotopy section of q. Repeating this for all j, we conclude that $\mathrm{secat}(q) \leq \mathrm{cat}_G(X)$. □

Definition 2.55. *The* contractible sectional category *of a map $p : E \to B$, denoted by* consecat(p)*, is the least integer k such that B may be covered by k contractible open sets U_j that each admit a map $s_j : U_j \to E$ where $p \circ s_j$ is homotopic to the inclusion $i_j : U_j \hookrightarrow B$.*

Remark 2.20. *Trivially,* $\mathrm{consecat}(p) \geq \mathrm{secat}(p)$ *for any map p.*

Proposition 2.22. *If X is a G-space such that the quotient map $q : X \to X/G$ is a fibration up to homotopy, then* $\mathrm{cat}_G(X) \leq \mathrm{consecat}(q)$.

Proof. Take $\mathrm{consecat}(q)$-many contractible open sets $U_j \subseteq X/G$ with $X/G = \cup_j U_j$ that each admit a homotopy section $s_j : U_j \to X$ of q. Define $V_j = q^{-1}[U_j]$, and note that by lemma 2.7 that each V_j is G-categorical. Clearly $X = \cup_j V_j$, so we conclude that $\mathrm{cat}_G(X) \leq \mathrm{consecat}(q)$. □

Question 2.1. *The homotopy sections in the proof above are completely unused. Can something stronger be proven?*

Question 2.2. *When do we have* $\mathrm{consecat}(p) = \mathrm{secat}(p)$*? Or are there any other bounds we can put on* $\mathrm{secat}(p)$ *via* $\mathrm{consecat}(p)$*?*

Recall that when $p : E \to B$ is a fibration we have that definitions 2.10 and 2.26 coincide (i.e. homotopy sections induce sections), so we stand to learn a lot about orbit canonicalizations if we can discern when $q : X \to X/G$ is a fibration.

Theorem 2.9. *This is part of theorem 1.8 in [Rai06]. A fiber bundle $p : E \to B$ over paracompact space B is a fibration.*

2.2.2 Bounds for Strong and Weak Sectional Category

We first introduce some (slightly) new terminology.

Definition 2.56. *The* weak sectional category *of a map $p : E \to B$, denoted* $\operatorname{secat}(p)$, *is the least integer k such that there k open sets U_j covering B that each admit a homotopy section $s_j : U_j \to E$ of p (i.e. $p \circ s_j \simeq i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion).*

Definition 2.57. *The* strong sectional category *of a map $p : E \to B$, denoted* $\operatorname{secat}^{\mathrm{s}}(p)$, *is the least integer k such that there k open sets U_j covering B that each admit a section $s_j : U_j \to E$ of p (i.e. $p \circ s_j = i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion).*

Recall that by remark 2.6 $\operatorname{secat}(p) = \operatorname{secat}^{\mathrm{s}}(p)$ when p is a fibration. In general, we have the following result:

Lemma 2.8. *For any map $p : E \to B$,* $\operatorname{secat}(p) \leq \operatorname{secat}^{\mathrm{s}}(p)$.

Proof. Take $\operatorname{secat}^{\mathrm{s}}(p)$-many open sets covering U_j with sections $s_j : U_j \to E$ of p i.e. $p \circ s_j = i_j$. It trivially follows that $p \circ s_j \simeq i_j$, so we are done. ☐

Definition 2.58. *(Definition 1.1 in [Coh26]) Let E, B, and F be spaces, and let $p : E \to B$ be a map such that*

1. *p is surjective,*
2. *for all $b \in B$, $p^{-1}[\{b\}] \cong F$,*
3. *and for all $b \in B$, there is some open neighborhood $U_b \subseteq B$ of B and a homeomorphism $\Phi_b : p^{-1}[U_b] \to U_b \times F$ such that $p|_{p^{-1}[U_b]} = \pi_b \circ \Phi_b$ where $\pi_b : U_b \times F \to U_b$ is the projection.*

Then p is a fiber bundle with fiber F.

Lemma 2.9. *(Corollary 2.7.14 in [Spa66]) If B is paracompact, then any fiber bundle $p : E \to B$ is a fibration.*

Definition 2.59. *(Definition 1.5 in [Coh26]) A topological group G is a* Lie group *is G is a differentiable manifold such that products and inverses are differentiable maps.*

Definition 2.60. *(Definition 1.6 in [Coh26]) Let G be a topological group, let E and B be spaces, and let $p : E \to B$ be a fiber bundle with fiber G such that*

1. *E has a free left G-action where $p(e) = p(ge)$ for all $g \in G$, $e \in E$, i.e. the fibers of p are closed under the G-action,*
2. *for all $b \in B$, the induced G-action on $p^{-1}[\{b\}]$ is free and transitive,*
3. *and for every open set $U \subseteq B$, there exists a G-equivariant homeomorphism $\Psi_U : p^{-1}[U] \to G \times U$ where G acts on $G \times U$ by multiplication on G.*

Then p is a principal G-bundle.

Definition 2.61. *(Definition 1.7 in [Coh26]) A G-space X* has slices *if the quotient map $q : X \to X/G$ has local sections around every point in X/G.*

Proposition 2.23. *(Proposition 1.2 in [Coh26]) If X is a free G-space with slices, then the quotient map $q : X \to X/G$ is a principal G-bundle.*

Theorem 2.10. *(Theorem 1.3 in [Coh26]) Let M be a smooth manifold with a smooth, free G-action by a compact Lie group G. Then M has slices.*

Corollary 2.10. *Let M be a smooth manifold with a smooth, free G-action by a compact Lie group G. Then $q : M \to M/G$ is a principal G-bundle.*

Proof. This follows directly from theorem 2.10 and proposition 2.23. □

Theorem 2.11. *Let M be a smooth manifold with a smooth, free G-action by a compact Lie group G such that M/G is paracompact. Then $q : M \to M/G$ is a fibration.*

Proof. This follows directly from corollary 2.10 and lemma 2.9. □

Theorem 2.12. *Let G be a compact Lie group and let M be a smooth manifold with submanifold $N \subseteq M$ such that*

1. *G acts smoothly on M,*
2. *N is G-invariant (hence $N/G \subseteq M/G$),*
3. *the induced G-action on N is free,*
4. *and N/G is paracompact.*

Then letting $q : M \to M/G$ and $p = q|_N : N \to N/G$ be the respective quotient maps and letting $p^* : H^*(N/G) \to H^*(N)$ be the induced map on cohomology, we have the following chain of inequalities:

$$\operatorname{nil}\ker(p^*) < \operatorname{secat}(p) = \operatorname{secat}^{\mathrm{s}}(p) \leq \operatorname{secat}^{\mathrm{s}}(q).$$

Proof. Since G is a compact Lie group acting freely on N and N/G is paracompact, by theorem 2.11 we know that p is a fibration. Then by proposition 2.6 we have that $\operatorname{nil}\ker(p^*) < \operatorname{secat}(p)$; by remark 2.6 we have that $\operatorname{secat}(p) = \operatorname{secat}^{\mathrm{s}}(p)$; and by lemma 2.2 we have that $\operatorname{secat}^{\mathrm{s}}(p) \leq \operatorname{secat}^{\mathrm{s}}(q)$. □

Theorem 2.12 gives a concrete method to find a lower bound for strong sectional category even for quotient maps that fail to be fibrations by considering appropriate submanifolds of a given space. This is particularly useful in the problem of orbit canonicalization, since orbit canonicalizations are directly related to the strong sectional category of the quotient map.

2.2.3 Canonicalizing $\mathbb{R}^{d\times n}$ with respect to $\mathrm{O}(d)$ and $\mathrm{SO}(d)$

Definition 2.62. *The* orthogonal group *of degree n is the set*

$$\mathrm{O}(n) := \left\{Q \in \mathbb{R}^{n\times n} : QQ^T = I\right\}.$$

Definition 2.63. *The* special orthogonal group *of degree n is the set*

$$\mathrm{SO}(n) := \left\{Q \in \mathbb{R}^{n\times n} : \det(Q) = 1\right\}.$$

Remark 2.21. *Observe that $\mathrm{SO}(n) \subseteq \mathrm{O}(n)$.*

Definition 2.64. *The set of* real symmetric matrices of degree n *is*

$$\mathbb{S}^n := \left\{A \in \mathbb{R}^{n\times n} : A = A^T\right\}.$$

Definition 2.65. *The set of* positive semi-definite matrices of degree n *is*

$$\mathbb{S}^n_+ := \left\{A \in \mathbb{S}^n : \forall x \in \mathbb{R}^n, x^T Ax \geq 0\right\}.$$

Definition 2.66. *The set of* negative semi-definite matrices of degree n *is*

$$\mathbb{S}^n_- := \left\{A \in \mathbb{S}^n : \forall x \in \mathbb{R}^n, x^T Ax \leq 0\right\}.$$

Definition 2.67. *The set of* positive semi-definite matrices of degree n with at most rank d *is*

$$\mathbb{S}^n_{\leq d} := \left\{A \in \mathbb{S}^n_+ : \mathrm{rank}(A) \leq d\right\}.$$

Remark 2.22. $\mathbb{S}^n_{\leq d} = \mathbb{S}^n_{\leq n}$ *for any $d \geq n$.*

Proposition 2.24. *$A \in \mathbb{S}^n_+$ if and only if there is some matrix $B \in \mathbb{R}^{n\times n}$ such that $A = B^T B$ [Wik26b].*

Corollary 2.11. $\mathbb{S}^n_+ = \mathbb{S}^n_{\leq n}$.

Definition 2.68. *The set of* positive semi-definite matrices of degree n with rank d *is*

$$\mathbb{S}^n_d := \left\{A \in \mathbb{S}^n_+ : \mathrm{rank}(A) = d\right\}.$$

Proposition 2.25. *$A \in \mathbb{S}^n_d$ if and only if there is some matrix $B \in \mathbb{R}^{d\times n}$ such that $A = B^T B$ with $\mathrm{rank}(B) = d$. [Wik26b].*

Lemma 2.10. *With $O(d)$ acting on $\mathbb{R}^{d\times n}$ via matrix multiplication on the left, $\mathbb{R}^{d\times n}/\,\mathrm{O}(d)$ is homeomorphic to the space $S := \left\{A^T A : A \in \mathbb{R}^{d\times n}\right\}$.*

Proof. To begin, consider the map $\phi : \mathbb{R}^{d\times n}$ given by $\phi(A) = A^T A$. Now let $A, B \in \mathbb{R}^{d\times n}$ such that $[A] = [B] \in \mathbb{R}^{d\times n}/O(d)$, i.e. there exists $Q \in O(d)$ such that $A = QB$. This means that $A^T A = (QB)^T QB = B^T Q^T QB = B^T B$, so ϕ induces a map $\widehat{\phi} : \mathbb{R}^{d\times n}/O(d) \to S$ via $\widehat{\phi}([A]) = A^T A$. Note that $\widehat{\phi}$ is bijective by construction, with inverse $\widehat{\phi}^{-1} : S \to \mathbb{R}^{d\times n}/O(d)$ given by $\widehat{\phi}^{-1}(A^T A) = [A]$. Both of these maps are continuous, so we are done. $\square$

Corollary 2.12. $\mathbb{R}^{d\times n}/O(d) \cong \mathbb{S}^n_{\leq d}$.

Proof. This follows directly from proposition 2.25 and lemma 2.10. $\square$

With corollary 2.12 in hand, we have that any invariant map $f : \mathbb{R}^{d\times n} \to Y$ factors through $\mathbb{S}^n_d$ and so it suffices to reduce to consider maps from $\mathbb{S}^n_d$ to Y.

Definition 2.69. *Given a group G and a G-set X, the* free locus *of X is*

$$X_{free} := \{x \in X : \forall g \in G \setminus \{e\}, gx \neq x\} = \{x \in X : G_x = \{e\}\}$$

Lemma 2.11. *With $O(d)$ acting on $\mathbb{R}^{d\times n}$ via matrix multiplication, $\mathbb{R}^{d\times n}_{free} = \mathbb{R}^{d\times n}_*$.*

Proof. First, let $A \in \mathbb{R}^{d\times n}$ such that $\mathrm{rank}(A) = k < d$. Then there are some orthonormal bases $\{e_1, ..., e_k\}$ of $\mathrm{col}(A)$ and $\{e_{k+1}, ..., e_d\}$ of $\mathrm{null}(A)$. Let $B = [e_1 \cdots e_d]^T \in O(d)$ be the change of basis matrix from the standard basis for $\mathbb{R}^d$ to $\{e_1, ..., e_d\}$. Then A can be written in the basis $\{e_1, ..., e_d\}$ as a matrix

$$\widehat{A} = BA = \begin{bmatrix} \widehat{a}_{1,1} & \cdots & \widehat{a}_{n,1} \\ \vdots & & \vdots \\ \widehat{a}_{k,1} & \cdots & \widehat{a}_{n,k} \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix}.$$

Now define the matrix $R \in \mathbb{R}^{d\times d}$ by $R_{i,i} = 1$ if $i \leq k$, $R_{i,i} = -1$ if $i > k$, and $R_{i,j} = 0$ if $i \neq j$ (R simply flips the sign of the last $d - k$ coordinates) and observe that clearly $R \in O(d)$. Define $Q = B^T RB$, and then $Q \in O(d)$ since

$$Q^T Q = (B^T RB)^T B^T RB = B^T R^T BB^T RB = I.$$

By construction we have that $R\widehat{A} = \widehat{A}$, so it follows that $QA = A$. However, without loss of generality $Q \neq I$, so we have that $A \notin \mathbb{R}^{d\times n}_{free}$.

Conversely, let $A \in \mathbb{R}^{d\times n}_*$ and let $Q \in O(d)$ such that $QA = A$. Since $\mathrm{col}(A) = \mathbb{R}^d$, this means that Q fixes all of $\mathbb{R}^d$ and hence $Q = I$. Therefore $A \in \mathbb{R}^{d\times n}_{free}$. $\square$

Corollary 2.13. *$O(d)$ does not act freely on any subset of $\mathbb{R}^{d\times n}$ if $n < d$.*

Corollary 2.14. $\mathbb{R}^{d\times n}_*/O(d) \cong \mathbb{S}^n_d$.

Proof. This follows from proposition 2.25, lemma 2.10, and lemma 2.11. $\square$

We now present a somewhat simplified proof of the result of [CG13] that $(\mathbb{R}^{d\times n}, O(d))$ and $(\mathbb{R}^{d\times n}, SO(d))$ lack continuous canonicalizations for $n > d$ and $n \geq d$ respectively. Propositions 2.26 and 2.27 and theorem 2.13 are due to Casson, as presented in theorem 1.1 of [Ken+99].

Definition 2.70. *For a finite dimensional matrix $A \in \mathbb{R}^{n \times m}$, the* spectrum *of A is the set of (possibly complex) eigenvalues of A, denoted $\sigma(A)$.*

Note: this is a slight abuse of the standard definition of the spectrum of a linear operator, but since we are in finite dimensions this definition coincides with the more general one.

Lemma 2.12. *Let $E \in \mathbb{S}^n \setminus \{0\}$ with $\operatorname{tr}(E) = 0$, and let $\varepsilon \in \mathbb{R}$ be the least eigenvalue of E. Then $|\varepsilon| > \frac{\|E\|}{n}$ where $\|\cdot\|$ denotes the operator norm.*

Proof. Assume for the sake of contradiction that $\varepsilon \leq \frac{\|E\|}{n}$. Let $\{\lambda_i\}$ and $\{\mu_j\}$ be the positive and negative eigenvalues of E respectively. Recall that $\operatorname{tr}(E) = \sum_i \lambda_i + \sum_j \mu_j$, so E must have at least one positive and one negative eigenvalue since $E \neq 0$. Also note that we must have that $|\mu_j| \leq |\varepsilon|$ for all j. From functional analysis we know that $\|E\| = \max\{|\min \sigma(E)|, |\max \sigma(E)|\}$, so since $|\varepsilon| < \|E\|$ we must have some positive eigenvalue $\lambda_j = \|E\|$. Then we have

$$\|E\| \leq \sum_i \lambda_i = \sum_j |\mu_j| \leq (n-1)|\varepsilon| < n|\varepsilon| \leq \|E\|$$

which is a contradiction. Therefore we conclude that $|\varepsilon| > \frac{\|E\|}{n}$. $\square$

Proposition 2.26. *Let $Y_+ = \mathbb{S}^n_+ \setminus \{0\}$ be the set of non-zero positive semidefinite matrices of degree n, let $Y_0 = \{A \in Y_+ : \det(A) = 0\}$, and let $\mathbb{R}_+$ be the group of positive real numbers under multiplication. If $\mathbb{R}_+$ acts on the left of Y_+ via scalar multiplication, then $Y_+/\mathbb{R}_+$ is homeomorphic to a closed ball of dimension $n(n+1)/2 - 1$. Additionally, $Y_0/\mathbb{R}_+$ is homeomorphic to the ball's boundary.*

Proof. The basic idea of this proof is that we observe that Y_+ is a convex cone without its point, so we simply take a slice of it and homeomorphically map that slice to a disk. Then it is reasonably clear that such a slice is exactly the cone quotient by scaling, so we conclude there. The difficulty lies in showing the slice is actually a topological disk, the details of which lie in this proof.

Let $\langle A, B\rangle := \operatorname{tr}(AB)$ be the Frobenius inner product on $\mathbb{S}^n$, which induces the Frobenius norm $\|\cdot\|_F$. Then let $I^\perp$ be orthogonal complement of I in $\mathbb{S}^n$, let $D := (I + I^\perp) \cap Y_+$, and let $B := \overline{B(0,1)} \cap I^\perp$ be the closed unit ball in $I^\perp$ under the operator, which has dimension $\dim(I^\perp) = \dim(Y_+) - \dim(\operatorname{span}\{I\}) = n(n+1)/2 - 1$. Observe that D is homeomorphic to $Y_+/\mathbb{R}_+$ via the quotient map $E \mapsto [E]$ with continuous inverse $[E] \mapsto \frac{1}{\|\operatorname{proj}_I E\|} E$.

Define the function $\mu : \mathbb{S}^n \to \mathbb{R}$ by $\mu(A) = \min \sigma(A)$. μ is well-defined since real symmetric matrices have finitely many eigenvalues, all of which are real, and it follows from Weyl's inequality that μ is continuous. For any $A \in \mathbb{S}^n$, observe that $I + A \in Y_+$ if and only if $\mu(I + A) \geq 0$.

We now define $\Phi : B \to D$ via $\Phi(E) = I - \frac{\|E\|}{\mu(E)} E$ if $E \neq 0$ and $\Phi(0) = I$. We will show that Φ is the desired homeomorphism. Observe that $\Phi(E) \in (I + I^\perp)$ by construction, but it remains to show that $\Phi(E) \in Y_+$. This is trivially true for $E = 0$, so we consider $E \neq 0$. Note that $E \in E^\perp$ means that $\operatorname{tr}(E) = 0$, meaning that the sum of E's eigenvalues is 0, so

E must have at least one negative eigenvalue since $E \neq 0$. Hence, $\mu(E) < 0$. Then we see that $E \in Y_+$ since

$$\begin{aligned}
\min_{\|x\|=1} x^T \Phi(E) x &= \min_{\|x\|=1} x^T \left(I - \frac{\|E\|}{\mu(E)} E \right) x \\
&= 1 - \frac{\|E\|}{\mu(E)} \min_{\|x\|=1} x^T E x \\
&= 1 - \|E\| \\
&\geq 0.
\end{aligned}$$

This also shows that $\Phi(E) \in Y_0$ if and only if $\|E\| = 1$. The continuity of Φ at every point other than 0 is evident from its construction. To see Φ's continuity at 0, first observe that

$$\frac{\mu(E)}{\|E\|} = \frac{\min_{\|x\|=1} x^T E x}{\|E\|} = \min_{\|x\|=1} x^T \frac{E}{\|E\|} x = \mu\left(\frac{E}{\|E\|}\right)$$

so

$$\Phi(E) = I - \frac{1}{\mu\left(\frac{E}{\|E\|}\right)} E.$$

Note that $\Phi(E)$ is continuous if and only if $\frac{1}{\mu\left(\frac{E}{\|E\|}\right)} E$ is, and since $\left\|\frac{E}{\|E\|}\right\| = 1$ we use lemma 2.12 to get that $\mu\left(\frac{E}{\|E\|}\right) > \frac{1}{n}$ and compute

$$\lim_{E \to 0} \left\| \frac{1}{\mu\left(\frac{E}{\|E\|}\right)} E \right\| < \lim_{E \to 0} \|nE\| = 0$$

so $\lim_{E \to 0} \Phi(E) = I$ and Φ is continuous.

We show that Φ is indeed a homeomorphism by constructing its inverse $\Phi^{-1} : D \to B$ via $\Phi^{-1}(I + E) = -\frac{\mu(E)}{\|E\|} E$ for $E \neq 0$ and $\Phi^{-1}(I) = 0$. To see that this is indeed the inverse of Φ, we first compute

$$\begin{aligned}
\Phi(\Phi^{-1}(I + E)) &= \Phi\left(-\frac{\mu(E)}{\|E\|} E \right) \\
&= I - \frac{\left\| -\frac{\mu(E)}{\|E\|} E \right\|}{\mu\left(-\frac{\mu(E)}{\|E\|} E \right)} \left(-\frac{\mu(E)}{\|E\|} E \right) \\
&= I + \frac{-\frac{\mu(E)}{\|E\|} \|E\|}{-\frac{\mu(E)}{\|E\|} \mu(E)} \frac{\mu(E)}{\|E\|} E \\
&= I + E
\end{aligned}$$

when $E \neq 0$, and the computation is trivial when $E = 0$. Similarly, we have

$$\begin{aligned}
\Phi^{-1}(\Phi(E)) &= \Phi^{-1}\left(I - \frac{\|E\|}{\mu(E)}E\right) \\
&= -\mu\left(\frac{-\frac{\|E\|}{\mu(E)}E}{\left\|-\frac{\|E\|}{\mu(E)}E\right\|}\right)\left(-\frac{\|E\|}{\mu(E)}E\right) \\
&= \frac{-\frac{\|E\|}{\mu(E)}}{-\frac{\|E\|}{\mu(E)}\|E\|}\mu(E)\frac{\|E\|}{\mu(E)}E \\
&= E.
\end{aligned}$$

Thus we have that Φ^{-1} is indeed the inverse of Φ. Lastly, Φ^{-1} is again clearly continuous everywhere except 0 by construction, and at 0 we have by the continuity of μ that

$$\lim_{E\to 0}\left\|\Phi^{-1}(E)\right\| = \lim_{E\to 0}\left\|-\frac{\mu(E)}{\|E\|}E\right\| = \lim_{E\to 0}|\mu(E)|\frac{\|E\|}{\|E\|} = 0.$$

We conclude that Φ is a homeomorphism, hence $Y_+/\mathbb{R}_+ \cong B$ with $Y_0/\mathbb{R}_+ \cong \partial B$. ☐

Alternative proof. First, we wish to show that the identity matrix I is interior point of $Y_+ \subseteq \mathbb{S}^n$ in the subspace topology. Recall all norms are equivalent on a finite dimensional vector space, so it suffices to find an open ball under the operator norm. Define the function $\mu : \mathbb{S}^n \to \mathbb{R}$ by $\mu(A) = \min \sigma(A)$. μ is well-defined since real symmetric matrices have finitely many eigenvalues, all of which are real, and it follows from Weyl's inequality that μ is continuous. For any $A \in \mathbb{S}^n$, observe that $I + A \in Y_+$ if and only if $\mu(I + A) \geq 0$. By the continuity of μ, there is some δ such that $\|A\| < \delta$ implies $|\mu(I) - \mu(I + A)| < 1$, meaning $\mu(I + A) > 0$ since $\mu(I) = 1$. Therefore $B(I, \delta) \subseteq Y_+$, as desired, so $\dim Y_+ = \dim \mathbb{S}^n = n(n + 1)/2$.

Next, Y_+ is clearly convex since the sum of any two positive semidefinite matrices is also positive semidefinite. From here, one can simply consider the intersection $Y_+ \cap (I + I^\perp)$ which is a bounded, convex subset of Y_+ and is hence homeomorphic to the ball of dimension $n(n + 1)/2 - 1$. Additionally, $\partial Y_+ = Y_0$ since any singular matrix can be perturbed to have a negative eigenvalue, so $Y_0 \cap (I + I^\perp)$ is homeomorphic to the ball's boundary. ☐

Proposition 2.27. *Let $X_+ = \{A \in \mathbb{R}^{n\times n} \setminus \{0\} : \det(A) \geq 0\}$ with left $\mathrm{SO}(n)$ action given by matrix multiplication, and let $Y_+ = \mathbb{S}^n_+ \setminus \{0\}$. Additionally, let $X_0 \subseteq X_+$ and $Y_0 \subseteq Y_+$ be the subspaces of singular matrices. Then $X_+/\mathrm{SO}(n) \cong Y_+$ with $X_0/\mathrm{SO}(n) \cong Y_0$.*

Proof. This follows from a nearly identical argument to that of lemma 2.10 and corollary 2.12, simply with the added care to only use matrices with nonnegative determinant. ☐

Theorem 2.13 (Casson)**.** *Let $X = \mathbb{R}^{n\times n} \setminus \{0\}$ with $\mathrm{SO}(n)$ and $\mathbb{R}_+$ acting on the left via matrix and scalar multiplication respectively, and let S be the sphere of dimension $n(n + 1)/2 - 1$. Then $X/(\mathbb{R}_+ \times \mathrm{SO}(n)) \cong S$.*

Proof. First observe that the actions of $\mathbb{R}_+$ and $\mathrm{SO}(n)$ on X commute, so we have that $X/(\mathbb{R}_+ \times \mathrm{SO}(n)) \cong (X/\mathrm{SO}(n))/\mathbb{R}_+$. Let X_0, X_+, X_- be the subspaces of X of matrices with zero, nonnegative, and nonpositive determinant respectively, and observe that clearly $X \cong X_+ \cup_{X_0} X_-$. Observe that any $\mathrm{SO}(n)$ orbit is entirely contained in X_0, X_+, or X_-, since $\mathrm{SO}(n)$-actions do not change the determinant. Additionally, $X_+ \cong X_-$ via your favorite map that flips the sign of the determinant (e.g. multiplying the last column by -1). It then follows that

$$(X/\mathrm{SO}(n))/\mathbb{R}_+ \cong (X_+/\mathrm{SO}(n))/\mathbb{R}_+ \cup_{(X_0/\mathrm{SO}(n))/\mathbb{R}_+} (X_+/\mathrm{SO}(n))/\mathbb{R}_+.$$

By propositions 2.26 and 2.27 we know that $(X_+/\mathrm{SO}(n))/\mathbb{R}_+$ and $(X_0/\mathrm{SO}(n))/\mathbb{R}_+$ are homeomorphic to the ball of dimension $n(n+1)/2-1$ and its boundary respectively, so we conclude that $X/(\mathbb{R}_+ \times \mathrm{SO}(n))$ is homeomorphic to the sphere of dimension $n(n+1)/2-1$. $\square$

With Casson's theorem in hand we can now rather quickly prove two of [DLS24]'s impossibility results.

Lemma 2.13. *(From [DLS24]) Let $G \subseteq \mathbb{R}^{d\times d}$ be a group acting on $\mathbb{R}^{d\times n}$ on the left via matrix multiplication. Then a canonicalization for $\mathbb{R}^{d\times n}$ induces a canonicalization for $\mathbb{R}^{d\times m}$ for any $m \leq n$.*

Proof. Let $f : \mathbb{R}^{d\times n} \to \mathbb{R}^{d\times n}$ be a canonicalization, and consider the map $\widehat{f} : \mathbb{R}^{d\times m} \to \mathbb{R}^{d\times m}$ given by $\widehat{f}(A) = f([A \mid 0])[:, : m]$ ($[A \mid 0]$ indicates padding A on the right with zeroes so $[A \mid 0] \in \mathbb{R}^{d\times n}$). Then for any $Q \in G$ and $A \in \mathbb{R}^{d\times m}$,

$$\widehat{f}(QA) = f(Q[A \mid 0])[:, : m] = (\widehat{Q}[A \mid 0])[:, : m] = \widehat{Q}A$$

for some $\widehat{Q} \in G$. Note that $\widehat{f}(QB) = \widehat{Q}A$ any $Q \in G$, and $B \in GA$ since f is a canonicalization, so we conclude that $\widehat{f}$ is a canonicalization as well. $\square$

Theorem 2.14. *If $n \geq d$, $\mathbb{R}^{d\times n}$ has no canonicalization with respect to $\mathrm{SO}(d)$.*

Proof. By lemma 2.13 we know that any canonicalization of $\mathbb{R}^{d\times n}$ induces a canonicalization for $\mathbb{R}^{d\times d}$, so we reduce to considering this case. Now, assume for the sake of contradiction that we have a canonicalization for $\mathbb{R}^{d\times d}$, and recall that this induces a section of the quotient map $\mathbb{R}^{d\times d} \to \mathbb{R}^{d\times d}/\mathrm{SO}(d)$. Observe that such a section also induces a section on any $\mathrm{SO}(d)$-invariant subspace of $\mathbb{R}^{d\times d}$, so we specifically consider the unit sphere $S^{d^2-1} \subseteq \mathbb{R}^{d\times d}$ under the Frobenius norm. It is easy to see that S^{d^2-1} is invariant under $\mathrm{SO}(d)$ transformations since orthogonal transformations do not change the Frobenius norm, and it is also clear that S^{d^2-1} is homeomorphic to the quotient $(\mathbb{R}^{d\times d} \setminus \{0\})/\mathbb{R}_+$ since one can always scale a non-zero matrix to have norm 1. Hence, we have a section $s : S^{d^2-1}/\mathrm{SO}(d) \to S^{d^2-1}$ of the quotient map $q : S^{d^2-1} \to S^{d^2-1}/\mathrm{SO}(d)$.

Recall by theorem 2.13 that the quotient $S^{d^2-1}/\mathrm{SO}(d)$ is homeomorphic to the sphere of dimension $d(d+1)/2-1$, $S^{\frac{d(d+1)}{2}-1}$. So, up to slight abuse of notation, we have induced maps on homology

$$H_k(S^{\frac{d(d+1)}{2}-1}) \xrightarrow{s_\sharp} H_k(S^{d^2-1}) \xrightarrow{q_\sharp} H_k(S^{\frac{d(d+1)}{2}-1})$$

such that $q_\sharp \circ s_\sharp$ is still the identity. However, when $k = d(d+1)/2 - 1$ this is impossible, since in this case $H_k(S^{\frac{d(d+1)}{2}-1}) \cong \mathbb{Z}$ and $H_k(S^{d^2-1}) \cong 0$. Therefore we conclude that there can be no canonicalization of $\mathbb{R}^{d\times n}$ with respect to $\mathrm{SO}(d)$ for $n \geq d$. $\square$

Theorem 2.15. *If $n > d$, $\mathbb{R}^{d\times n}$ has no canonicalization with respect to* $\mathrm{O}(d)$*.*

Proof 1. This proof proceeds almost identically to that of theorem 2.14. We first reduce to considering $\mathbb{R}^{d\times(d+1)}$, then find $S^{d(d+1)-1} \subseteq \mathbb{R}^{d\times(d+1)}$, which is homeomorphic to $\mathbb{R}^{d\times(d+1)}/\mathbb{R}_+$. Next, observe that $S^{d(d+1)-1}/O(d)$ is homeomorphic to $\mathbb{S}^{d+1}_{\leq d}/\mathbb{R}_+$ by corollary 2.12, which is in turn homeomorphic to the boundary of the ball of dimension $(d+1)(d+2)/2-1$, i.e. the sphere of one dimension less than that, by proposition 2.26. Then a contradiction once again arises by considering the induced maps on homology. We require $d \geq 4$ here so that $d^2 - 1 > (d+1)(d+2)/2 - 2$ and this last step actually produces a contradiction. For lower values of d, one can use [DLS24]'s proof. $\square$

This result also has a more direct proof.

Proof 2. Assume for the sake of contradiction that we have some canonicalization of $\mathbb{R}^{d\times n}$ and the associated section $s : \mathbb{S}^n_{\leq d} \to \mathbb{R}^{d\times n}$. $\square$

Proposition 2.28. *Let* $\mathrm{SO}(d)$ *act on the left of $\mathbb{R}^{d\times d}_*$ via matrix multiplication and let $q : \mathbb{R}^{d\times d}_* \to \mathbb{R}^{d\times d}_*/$* $\mathrm{SO}(d)$ *be the quotient map. Then* $\mathrm{secat}^{\mathrm{s}}(q) = 2$.

Proof. First, observe that by theorem 2.14, proposition 2.15, remark 2.3, and lemma 2.2 we have that $\mathrm{secat}^{\mathrm{s}}(q) > 1$. Hence, it remains to show that $\mathrm{secat}^{\mathrm{s}}(q) \leq 2$. By proposition 2.25 we know that $\mathbb{R}^{d\times d}$ is homeomorphic to the (disjoint) union of full rank matrices in $\mathbb{S}^d_+$ and $\mathbb{S}^d_-$ (here denoted S_+ and S_-) via the map $q : \mathbb{R}^{d\times d}_* \to S_+ \sqcup S_-$ defined as follows:

$$q(A) = \begin{cases} A^T A & \text{if } \det(A) > 0 \\ -A^T A & \text{if } \det(A) < 0 \end{cases}.$$

Observe that this well-defined since $A^T A$ is always positive semi-definite with $\mathrm{rank}(A^T A) = \mathrm{rank}(A)$, so $A^T A \in S_+$ and $-A^T A \in S_-$. Additionally, q is clearly continuous.

Next, we define $s_+ : S_+ \to \mathbb{R}^{d\times d}$ by $s_+(A) = A^{\frac{1}{2}}$ and $s_- : S_- \to \mathbb{R}^{d\times d}$ by $s_-(A) = T(-A)^{\frac{1}{2}}$, where $M^{\frac{1}{2}}$ denotes the matrix square root where T is some orthogonal transformation that flips the sign of its input's determinant, i.e. $T \in O(d) \setminus SO(d)$. Both of these maps are clearly well-defined and continuous by construction. For $A \in S_+$, we then compute

$$q(s_+(A)) = q(A^{\frac{1}{2}}) = (A^{\frac{1}{2}})^T (A^{\frac{1}{2}}) = A$$

and similarly for $A \in S_-$ we have

$$q(s_-(A)) = q(T(-A)^{\frac{1}{2}}) = -(T(-A)^{\frac{1}{2}})^T (T(-A)^{\frac{1}{2}}) = -(-A)^{\frac{1}{2}} T^T T (-A)^{\frac{1}{2}} = A.$$

Therefore both s_+ and s_- are local sections of q one S_+ and S_- respectively, and both S_+ and S_- are open in the quotient by construction so we conclude that $\mathrm{secat}^{\mathrm{s}}(q) \leq 2$. $\square$

We would also like to compute or at least bound the sectional category of the quotients $\mathbb{R}^{d\times n} \to \mathbb{R}^{d\times n}/\operatorname{SO}(d)$ and $\mathbb{R}^{d\times n}/\operatorname{O}(d)$ in general. To do this, it is useful to consider the Stiefel and Grassmann manifolds as in section 2.1.4.

Remark 2.23. *(From [BZA24]) The Grassmann manifold* $\operatorname{Gr}(n,k)$ *can also be realized as a quotient of the Stiefel manifold* $\operatorname{St}(n,k)$ *by relating matrices that are basis for the same subspace, i.e. those that differ by an orthogonal transformation. This is captured by the action of* $O(k)$ *on* $\operatorname{St}(n,k)$ *by right multiplication, hence* $\operatorname{Gr}(n,k) \cong \operatorname{St}(n,k)/O(k)$.

Both of the the Stiefel and Grassman manifolds are compact, and have corresponding non-compact variants:

Definition 2.71. *The* non-compact Stiefel manifold $\operatorname{St}'(n,k)$ *is the set of basis matrices for* k*-dimensional subspaces of* $\mathbb{R}^n$*, which can be represented via*

$$\operatorname{St}'(n,k) := \left\{U \in \mathbb{R}^{n\times k} : \operatorname{rank}(U) = k\right\}$$

Definition 2.72. *The* non-compact Grassmann manifold $\operatorname{Gr}'(n,k)$ *is* $\operatorname{St}'(n,k)/O(k)$ *where* $O(k)$ *acts on the right of* $\operatorname{St}'(n,k)$ *via matrix multiplication.*

Remark 2.24. *The transposition map* $A \mapsto A^T$ *gives a homeomorphism between* $\mathbb{R}^{d\times n}_*$ *and* $\operatorname{St}'(n,d)$*, hence* $\mathbb{R}^{d\times n}_*/O(d) \cong \operatorname{Gr}'(n,d)$.

In view of remark 2.24, we now wish to compute/bound the sectional category of the projection $\pi : \operatorname{St}(n,k) \to \operatorname{Gr}(n,k)$, treating $\operatorname{St}(n,k)$ as the unit sphere in $\mathbb{R}^{d\times n}_*$. But recall that by corollary 2.8 this can be done by computing/bounding the cup length of $\operatorname{Gr}(n,k)$.

Proposition 2.29. *(From [Pav21] following proposition 3.5) For* $n \le m$*,* $\operatorname{cup}(\operatorname{Gr}(n,k)) \le \operatorname{cup}(\operatorname{Gr}(m,k))$.

Proposition 2.30. *(From [Pav21], proposition 3.3) If* $2^s < n \le 2^{s+1}$ *for some integer* s*, then*

$$\operatorname{cup}(\operatorname{Gr}(n,2)) = \begin{cases} 2^{s+1} - 2 & \text{if } n = 2^s + 1 \\ 2^{s+1} - 1 & \text{otherwise.} \end{cases}$$

Proposition 2.31. *(From [Sto82], p. 104) If* $2^s < n \le 2^{s+1}$ *for some positive integer* s*, we have*

$$\operatorname{cup}(\operatorname{Gr}(n,3)) = \begin{cases} 2^{s+2} - 3\cdot 2^{p-1} - 4 & \text{if } n = 2^{s+1} - 2^p + 1 \text{ for some } p \ge 1 \\ 2^{s+2} - 3\cdot 2^{p-1} - 4 + t & \text{if } n = 2^{s+1} - 2^p + t \\ & \quad \text{for some } p \ge 1, 2 \le t \le 2^{p-1} \\ 2^{s+2} - 5 & \text{if } n = 2^{s+1} \end{cases}$$

From this, we have a tight lower bound of $\operatorname{cup}(\operatorname{Gr}(n,3)) \ge 2n - 5$.

Proposition 2.32. *Letting* s *range over the positive integers, we have*

$$\operatorname{cup}(\operatorname{Gr}(n,4)) = \begin{cases} 3\cdot 2^s - 7 + 2^{r+1} + t & \text{if } n = 2^s + 1 + 2^r + t \\ & \quad \text{for some } 0 \le r < s, 0 \le t < 2^r \\ 3\cdot 2^s - 7 & \text{if } n = 2^s + 1 \end{cases}$$

From this, we have a tight lower bound of $\operatorname{cup}(\operatorname{Gr}(n,4)) \ge \frac{9}{4}n - 8$.

Proposition 2.33. *(From [Sto82], p.104) For $s \geq 3$, $\operatorname{cup}(\operatorname{Gr}(2^s+1,5)) = 3 \cdot 2^s + 2^{s-2} - 9$.*

Corollary 2.15. *Combining propositions 2.33 and 2.29, we have that*

$$\operatorname{cup}(\operatorname{Gr}(n,5)) \geq \frac{13}{8}n - \frac{85}{8}$$

For more general k, we have the following bound.

Proposition 2.34. *(Corollary of [Sto82], p. 103) If $2 \leq k \leq n/2$ and $2^s < n \leq s^{s+1}$ for some integer s, then*

$$\operatorname{cup}(\operatorname{Gr}(n,k)) \geq \begin{cases} 2^{s+1}-2 & \text{if } n = 2^s+1 \\ 2^{s+1}-1 & \text{otherwise.} \end{cases}$$

In turn, this means that $\operatorname{cup}(\operatorname{Gr}(n,k)) \geq n-1$.

All of these cup length bounds can thus be used to compute lower bounds for $\operatorname{secat}^{\mathrm{s}}(q)$ with the quotient map $q : \mathbb{R}^{d\times n} \to \mathbb{R}^{d\times n}/\operatorname{O}(d)$, summarized in the following theorem.

Theorem 2.16. *Let $n \geq d$ and let $q : \mathbb{R}^{d\times n} \to \mathbb{R}^{d\times n}/\operatorname{O}(d)$ be the quotient map. Then*

$$\operatorname{secat}(q) \geq \begin{cases} 2n-5 & \text{if } d=3 \\ \frac{9}{4}n - 8 & \text{if } d=4 \\ \frac{13}{8}n - \frac{85}{8} & \text{if } d=5 \\ n-1 & \text{if } 6 \leq d \leq \frac{n}{2}. \end{cases}$$

We now turn to the case of $q : \mathbb{R}^{d\times n} \to \mathbb{R}^{d\times n}/\operatorname{SO}(d)$. This case turns out to me markedly more difficult to study, as we now must consider the *oriented* Grassmann manifolds, which less is known about.

Definition 2.73. *The* oriented Grassmann manifold $\widetilde{\operatorname{Gr}}(n,k)$ *is the space of all oriented k-dimensional vector subspaces of $\mathbb{R}^n$.*

Remark 2.25. $\widetilde{\operatorname{Gr}}(n,k) \simeq \mathbb{R}^{d\times n}_*/\operatorname{SO}(d)$.

Remark 2.26. *There is a natural vector bundle over $\widetilde{\operatorname{Gr}}(n,k)$ given by*

$$\widetilde{E}(n,k) := \left\{(P,x) \in \widetilde{\operatorname{Gr}}(n,k) \times \mathbb{R}^n : x \in P\right\}^4$$

with the projection $p : \widetilde{E}(n,k) \to \widetilde{\operatorname{Gr}}(n,k)$.

Remark 2.27. *There is a natural projection $\widetilde{\pi} : \operatorname{St}(n,k) \to \widetilde{\operatorname{Gr}}(n,k)$ where $\widetilde{\pi}(B)$ is the plane* $\operatorname{span}(B)$ *with orientation given by the order of the basis B.*

[4] "$x \in P$" is a slight abuse of notation here, since P is not a set but rather an oriented plane. However, the meaning is still the obvious one: that x lies in the (unoriented) plane specified by P.

Proposition 2.35. *Let $\widetilde{\pi} : \mathrm{St}(n,k) \to \widetilde{\mathrm{Gr}}(n,k)$ be the natural projection. Then the induced map $\widetilde{\pi}^* : H^*(\widetilde{\mathrm{Gr}}(n,k)) \to H^*(\mathrm{St}(n,k))$ is zero on the Stiefel-Whitney classes $w_i(\widetilde{E}(n,k))$.*

Proof. This argument is exactly the same as proposition 2.18, except that we no longer have that $H^*(\widetilde{\mathrm{Gr}}(n,k);\mathbb{Z}_2)$ is generated by Stiefel-Whitney classes. $\square$

Definition 2.74. *The* height *of a cohomology class $x \in H^*(X)$ is*

$$ht(x) := \sup\left\{m \in \mathbb{Z}^+ : x^m \neq 0\right\}$$

where x^m denotes $\underbrace{x \smile x \smile \cdots \smile x}_{m \text{ times}}$.

Remark 2.28. *In light of proposition 2.35, we have that*

$$\operatorname{nil}\ker\widetilde{\pi}^* \geq \sup_{x \in H^*(\widetilde{\mathrm{Gr}}(n,k))} ht(x).$$

Theorem 2.17. *Let $n/2 \geq d \geq 3$ and let $q : \mathbb{R}^{d\times n} \to \mathbb{R}^{d\times n}/\,\mathrm{SO}(d)$ be the quotient map. Then*

$$\operatorname{secat}(q) \geq \begin{cases} \frac{n-d+6}{2} & \text{if } n-d+3 \geq 7 \text{ is odd and } n-d+3 \neq 9, 11 \\ \frac{n-d+5}{2} & \text{if } n-d+3 \geq 7 \text{ is even and } n-d+3 \neq 10, 12 \\ 5 & \text{if } n-d+3 = 9, 10, 11, 12. \end{cases}$$

Proof. This follows directly from remark 2.28 and the proof of Proposition B in [Kor06], where the author computes the heights of some Stiefel-Whitney classes. $\square$

We might also wish to giver an upper bound for the sectional category of the two quotient maps considered above. Unfortunately, I could not find a way to do this.

2.2.4 Canonicalizing $\mathbb{R}^{d\times n}$ with respect to Σ_n

With an extensive exploration of canonicalizing $\mathbb{R}^{d\times n}$ with respect to rotation, we now follow the lead of [DLS24] and consider canonicalizing $\mathbb{R}^{d\times n}$ with respect to permutations. More precisely, we wish to canonicalize $\mathbb{R}^{d\times n}$ with respect to Σ_n, the symmetric group on n elements. Practically speaking, this would allow us to operate on unordered collections of points in $\mathbb{R}^d$.

Definition 2.75. *The* symmetric group on n elements, *denoted Σ_n, is the set of all bijections on the first n positive integers with the product given by function composition.*

Definition 2.76. *For a topological space X and positive integer n,*

$$\mathrm{Conf}_n(X) := \{(x_1, ..., x_n) \in X^n : x_i \neq x_j \text{ for } i \neq j\}$$

is the space of n ordered mutually distinct points in X.

Definition 2.77. *For a topological space X and positive integer n,*

$$\mathrm{UConf}_n(X) := \mathrm{Conf}_n(X)/\Sigma_n = \{\{x_1, ..., x_n\} \subseteq X : x_i \neq x_j \text{ for } i \neq j\}$$

is the space of n unordered mutually distinct points in X.

Definition 2.78. *A G-action on X is* properly discontinuous *if for every $x \in X$ there is an open neighborhood $U \ni x$ such that $gU \cap U = \varnothing$ for all $g \in G$.*

Lemma 2.14. *Let G be a finite group that acts freely on a Hausdorff space X. Then the action of G is properly discontinuous.*

Proof. Let $x \in X$ and write $G = \{g_1, ..., g_n\}$ where $g_1 = e_G$. Since X is Hausdorff, we can find open neighborhoods $U_1, ..., U_n$ of $g_1x, ..., g_nx$ such that $U_i \cap U_j = \varnothing$ for $i \neq j$. [5] Now let $U = \cap_{i=2}^n g_i^{-1} U_i$ and observe that U is an open set containing x with $g_iU \subseteq U_i$. To finish, let $g \in G$ such that $gU \cap U \neq \varnothing$. Since $g_iU \cap U = \varnothing$ for $i \neq 1$ by construction, we conclude that $g = g_1$ and G acts properly discontinuously on X. □

Lemma 2.15. *If X is Hausdorff and $\mathrm{Conf}_n(X)$ is path-connected, then the quotient map $q : \mathrm{Conf}_n(X) \to \mathrm{UConf}_n(X)$ is a covering space. Additionally, q is an $n!$-sheeted covering.*

Proof. By proposition 1.40 of [Hat05], it suffices to show that the action of Σ_n on $\mathrm{Conf}_n(X)$ is properly discontinuous, so by lemma 2.14 it suffices to show that Σ_n acts freely on $\mathrm{Conf}_n(X)$. However, Σ_n clearly acts freely on $\mathrm{Conf}_n(X)$ by the definition of $\mathrm{Conf}_n(X)$, so q is a covering map.

For the second statement, proposition 1.40 of [Hat05] also gives us that

$$\Sigma_n \cong \pi_1(\mathrm{UConf}_n(X))/q_*(\pi_1(\mathrm{Conf}_n(X))),$$

so by proposition 1.32 of [Hat05] we conclude that the covering has card $(\Sigma_n) = n!$ sheets. □

[5] Got construction of U here

Remark 2.29. *Observe that* $\mathrm{Conf}_n(\mathbb{R}^d) \subseteq (\mathbb{R}^d)^n \cong \mathbb{R}^{d\times n}$ *and consequently* $\mathrm{UConf}_n(\mathbb{R}^d) \subseteq (\mathbb{R}^d)^n/\Sigma_n \cong \mathbb{R}^{d\times n}/\Sigma_n$.

Lemma 2.16. *For* $d \geq 2$, $\mathrm{Conf}_n(\mathbb{R}^d)$ *is path-connected.*

Proof. Let $x, y \in \mathrm{Conf}_n(\mathbb{R}^d)$ and write $x = (x_1, ..., x_n), y = (y_1, ..., y_n)$. Then for every i, we can find some path

$$\alpha_i : I \to \mathbb{R}^d \setminus \{y_1, ..., y_{i-1}, x_{i+1}, ..., x_n\}$$

such that $\alpha_i(0) = x_i$ and $\alpha_i(1) = y_i$ since $\mathbb{R}^d \setminus A$ for any finite set A is still path-connected. Then we simply construct a path $\gamma : I \to \mathrm{Conf}_n(\mathbb{R}^d)$ with $\gamma(0) = x$ and $\gamma(1) = y$ by moving coordinate-wise via the α_i's, i.e.

$$\gamma_i(t) = \begin{cases} x_i & \text{if } 0 \leq t < \frac{i}{n} \\ \alpha_i(nt - i) & \text{if } \frac{i}{n} \leq t < \frac{i+1}{n} \\ y_i & \text{if } \frac{i+1}{n} \leq t \leq 1. \end{cases}$$

We can see that $\gamma(t) \in \mathrm{Conf}_n(\mathbb{R}^d)$ for all $t \in [0, 1]$ since the α_i's are chosen such that when $\frac{i}{n} \leq t < \frac{i+1}{n}$, we have $\gamma_j(t) = y_j$ for $j < i$, $\gamma_k(t) = x_k$ for $k > i$, and $\gamma_i(t) = \alpha_i(nt - i) \notin \{y_1, ..., y_{i-1}, x_{i+1}, ..., x_n\}$. Therefore we conclude that $\mathrm{Conf}_n(\mathbb{R}^d)$ is path-connected. □

Remark 2.30. *Lemma 2.16 fails for* $\mathbb{R}$ *since removing any points from* $\mathbb{R}$ *causes it to no longer be path-connected. In greater generality, lemma 2.16 holds for any manifold* X *with* $\dim X \geq 2$, *and fails if* $\dim X = 1$ *for the same reasons as for* $\mathbb{R}$.

Lemma 2.17. *Let* $p : \widetilde{X} \to X$ *be a covering space where* $\widetilde{X}$ *is not homeomorphic to* X *(i.e. the covering space has more than one sheet). Then there is no global section of* p.

Proof. Assume for the sake of contradiction that we have a section $s : X \to \widetilde{X}$ of p, and fix some base point $x \in X$. Since p has more than one sheet, we can find $x_1 \in p^{-1}(x)$ that is distinct from the base point $x_0 \in \widetilde{X}$. Then by path-connectivity we can find a path $\gamma : I \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Since $p(x_0) = p(x_1)$ by construction, we have that $p \circ \gamma$ is a loop in X and $[p \circ \gamma]$ is a well-defined element of $\pi_1(X)$. Now observe that p and s induce maps $p_* : \pi_1(\widetilde{X}) \to \pi_1(X)$ and $s_* : \pi_1(X) \to \pi_1(\widetilde{X})$ such that $p_* \circ s_* = \mathrm{id}$. This means we can find $[\alpha] := s_*([p \circ \gamma]) \in \pi_1(\widetilde{X})$ such that $p_*([\alpha]) = [p \circ \alpha] = [p \circ \gamma]$, giving us a homotopy $h_t : I \to X$ such that $h_0 = p \circ \alpha$ and $h_1 = p \circ \underset{\sim}{\gamma}$. Since $\underset{\sim}{\gamma}$ clearly lifts $p \circ \gamma$, the homotopy lifting property gives us a unique homotopy lift $\widetilde{h}_t : I \to \widetilde{X}$ such that $p \circ \widetilde{h}_t = h_t$ with $\widetilde{h}_1 = \gamma$. By the uniqueness of lifting we also have that $\alpha = \widetilde{h}_0$, but this is impossible since α is a loop and γ is not. Therefore we conclude that there can be no global section s of p. □

We now present a new proof of the uncanonicalizability of $\mathbb{R}^{d\times n}$ with respect to Σ_n. [ZHP20] proved this using a metric argument, but observe that the proof of theorem 2.18 generalizes to any manifold of dimension at least 2.

Theorem 2.18. *If* $d, n \geq 2$, *there exists no continuous canonicalization of* $\mathbb{R}^{d\times n}$ *with respect to* Σ_n.

Proof. Since $d, n \geq 2$, we immediately conclude by lemmas 2.15, 2.16, and 2.17, as any canonicalization of $\mathbb{R}^{d\times n}$ must descend to a canonicalization of $\mathrm{Conf}_n(\mathbb{R}^d)$. $\square$

Remark 2.31. *For completeness, we note that for $d = 1$, sorting is a canonicalization, and for $n = 1$ the identity is a canonicalization.*

As per with rotations, we would like to give some lower bounds for the sectional category of the relevant quotient map here.

Theorem 2.19. *(Theorem 1.4 of [Rot08]) Let $q : \mathrm{Conf}_n(\mathbb{R}^d) \to \mathrm{Conf}_n(\mathbb{R}^d)/\Sigma_n$ be the quotient map, and let $\alpha(n)$ be the number of 1's in the dyadic expansion of n. Then we have*

$$(n - \alpha(n))(d - 1) + 1 \leq \mathrm{secat}(q) \leq \mathrm{cat}(\mathrm{Conf}_n(\mathbb{R}^d)/\Sigma_n) \leq (n - 1)(d - 1) + 1.$$

Corollary 2.16. *Let $q : \mathrm{Conf}_n(\mathbb{R}^d) \to \mathrm{Conf}_n(\mathbb{R}^d)/\Sigma_n$ be the quotient map. Then*

$$\mathrm{secat}(q) \geq (n - \lceil \log_2(n) \rceil)(d - 1) + 1$$

where $\lceil x \rceil$ is the least integer greater than or equal to x.

By theorem 2.18 we cannot find a continuous canonicalization of $\mathbb{R}^{d\times n}$ with respect to Σ_n, but theorem 2.19 tells us that, if we are willing to restrict to $\mathrm{Conf}_n(\mathbb{R}^d)$ we must be able to find a canonicalization with at most $(n-1)(d-1)+1$ open sets on which the canonicalization is continuous. Unfortunately, the proof of this fact does not appear to be constructive, so the question remains how to produce such a canonicalization that could be used in practice.

2.2.5 Canonicalizing $\mathbb{R}^{d\times n}/\Sigma_n$ with respect to $\mathrm{O}(d)$ and $\mathrm{SO}(d)$

The last space we might wish to canonicalize is unordered point clouds with respect to rotations. In particular, we wish to produce a section of the quotient $q : \mathrm{UConf}_n(\mathbb{R}^d) \to \mathrm{UConf}_n(\mathbb{R}^d)/O(d)$, or show that such a map cannot exist. We begin with a few observations.

Lemma 2.18. *The actions of Σ_n and $O(d)$ on $R^{d\times n}$ commute, and we have*

$$(\mathbb{R}^{d\times n}/O(d))/\Sigma_n \cong (\mathbb{R}^{d\times n}/\Sigma_n)/O(d) \cong \mathbb{R}^{d\times n}/(\Sigma_n \times O(d)).$$

Proof. This follows directly from the associativity of matrix multiplication. Explicitly, let $Q \in O(d)$ and $\sigma \in \Sigma_n$, treating σ as an n by n matrix. Then for any $A \in \mathbb{R}^{d\times n}$,

$$\sigma \cdot Q \cdot A = \sigma \cdot (QA) = (QA)\sigma = Q(A\sigma) = Q \cdot (A\sigma) = Q \cdot \sigma \cdot A.$$

□

Lemma 2.19. *The quotient $q : \mathrm{Conf}_n(\mathbb{R}^d)/O(d) \to (\mathrm{Conf}_n(\mathbb{R}^d)/O(d))/\Sigma_n$ is an $n!$-sheeted covering space.*

Proof. Recall by lemma 2.14 that it suffices to show that Σ_n acts freely on $\mathrm{Conf}_n(\mathbb{R}^d)/O(d)$. So, let $O(d)A, O(d)B \in \mathrm{Conf}_n(\mathbb{R}^d)/O(d)$ and $\sigma \in \Sigma_n$ such that $O(d)A = (O(d)B)\sigma$. But the actions of $O(d)$ and Σ_n commute so we have $(O(d)B)\sigma = O(d)(B\sigma)$, and we can find $Q \in O(d)$ such that $A = QB\sigma$. Since Σ_n acts freely on $\mathrm{Conf}_n(\mathbb{R}^d)$, we have that $A = QB$ We therefore conclude that $O(d)A = O(d)B$ and q is an $n!$-sheeted covering space. □

Theorem 2.20. *The quotient $q : \mathrm{Conf}_n(\mathbb{R}^d)/O(d) \to (\mathrm{Conf}_n(\mathbb{R}^d)/O(d))/\Sigma_n$ has no canonicalization.*

Proof. This follows directly from lemmas 2.17 and 2.19. □

Corollary 2.17. *The quotient $Q : \mathbb{R}^{d\times n} \to \mathbb{R}^{d\times n}/(O(d) \times \Sigma_n)$ has no canonicalization.*

Proof. This follows from theorem 2.20 and the fact that any section $S : \mathbb{R}^{d\times n}/(O(d) \times \Sigma_n) \to \mathbb{R}^{d\times n}$ induces a section of $q : \mathbb{R}^{d\times n}/O(d) \to (\mathbb{R}^{d\times n}/O(d))/\Sigma_n$ via $s := p \circ S$ where $p : \mathbb{R}^{d\times n} \to \mathbb{R}^{d\times n}/O(d)$ is the quotient map. □

Note that all the preceding results in this section apply just the same for $\mathrm{SO}(d)$ instead of $\mathrm{O}(d)$.

Chapter 3

Frame Averaging

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